Coláiste na hOllscoile Corcaigh
University College Cork, Ireland

MS4090 — Final Report

Hierarchical Machine Learning for Variance Risk Premium Estimation: From VIX Forecasting to Options Trading

Submitted by:	Andrew Fouhy
Student number:	122310513
Supervisor:	Dr. Mingchuan Zhao
Date:	March 2026

Abstract

The variance risk premium is the persistent wedge between implied and realised volatility. It represents one of the most economically significant risk premia in financial markets. This dissertation develops a hierarchical machine learning framework for estimating and exploiting the variance risk premium through systematic options trading on the S&P 500. Rather than forecasting the VRP directly, the proposed architecture decomposes the problem into two parallel forecasting tasks, next-day VIX prediction using XGBoost regression on AR(1) residuals, and 22-day-ahead realised volatility prediction using a separate XGBoost model trained on OHLC data. The VRP forecast is then constructed by combining these component predictions, enabling transparent error attribution and modular model improvement.

Using daily data spanning January 1990 to February 2026, which encompasses the dot-com bubble, the Global Financial Crisis, and the COVID-19 pandemic this study evaluates the XGBoost framework against standard econometric models including HAR and GARCH models.

For realised volatility, a comparison of five daily variance estimators including close-to-close, Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang demonstrates that range-based estimators achieve up to 2.46 times the statistical efficiency of close-to-close returns.

The dissertation contributes an AR(1)-residual XGBoost framing that isolates the nonlinear signal from the dominant autoregressive component, a multi-estimator realised volatility pipeline, a hierarchical VRP prediction architecture with formal error decomposition, and comprehensive regime-conditional evaluation across low, medium, and high volatility environments. The framework provides a foundation for systematic VRP harvesting strategies in the S&P 500 options market, while highlighting the fundamental challenge of horizon mismatch in combining one-step-ahead VIX forecasts with multi-step RV predictions.

Contents

List of Figures

List of Tables

List of Abbreviations

Abbreviation	Description
ADF	Augmented Dickey-Fuller test
AR	Autoregressive
AUC	Area Under the Curve
CBOE	Chicago Board Options Exchange
DM	Diebold-Mariano test
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
GBM	Geometric Brownian Motion
GK	Garman-Klass
HAC	Heteroskedasticity and Autocorrelation Consistent
HAR	Heterogeneous Autoregressive
IV	Implied Volatility
KPSS	Kwiatkowski-Phillips-Schmidt-Shin test
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MCS	Model Confidence Set
ML	Machine Learning
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
OHLC	Open-High-Low-Close
PP	Phillips-Perron test
QLIKE	Quasi-Likelihood loss
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
RS	Rogers-Satchell
RV	Realised Volatility
SDF	Stochastic Discount Factor
SHAP	SHapley Additive exPlanations
SPX	S&P 500 Index
VIX	CBOE Volatility Index
VRP	Variance Risk Premium
XGBoost	eXtreme Gradient Boosting
YZ	Yang-Zhang

Chapter 1

Introduction

1.1 Motivation

The variance risk premium, defined as the difference between the risk-neutral expected variance extracted from option prices and the physical expectation of future realised variance represents the compensation that investors demand for bearing the risk of future volatility fluctuations. Implied variance systematically exceeds realised variance reflecting a fundamental asymmetry in investor preferences. Market participants are willing to pay a premium for insurance against variance increases.

Extracting this premium is not straightforward. The VRP is the comparison between a forward looking implied measure and a backward looking physical measure. It follows it must be estimated from those two constituent components with each presenting its own measurement and forecasting challenges. Implied volatility, proxied by the CBOE Volatility Index (VIX) is the risk-neutral expectation. Realised volatility is estimated from historical price data, and the choice of estimator from close-to-close squared returns to range-based methods using OHLC data materially affects the resulting VRP series. Moreover, both components exhibit nonlinear dynamics, regime dependence, and time-varying persistence that challenge traditional linear econometric models.

Forecasting volatility is a well explored problem in financial econometrics. The GARCH family of models, originating with Engle (1982) and Bollerslev (1986), captures volatility clustering through an autoregressive structure in conditional variance. The Heterogeneous Autoregressive (HAR) model of Corsi (2009) decomposes volatility persistence across daily, weekly, and monthly horizons, motivated by the Heterogeneous Market Hypothesis. These models remain the dominant benchmarks against which new forecasting methods are evaluated.

The application of machine learning to volatility forecasting has accelerated substantially over the past decade. Christensen et al. (2023) demonstrated that gradient boosting and neural networks are often superior to time-series models, with improvements most pronounced at longer horizons. Gu et al. (2020) established that tree-based methods and neural networks approximately double the Sharpe ratios of leading regression-based strategies for equity risk premium measurement.

Yet relatively little work has applied these methods specifically to the hierarchical estimation of the variance risk premium through forecasting the VIX and realised volatility separately with machine learning, combining the forecasts to construct a VRP estimate, and then evaluating the resulting signals for systematic trading.

This dissertation addresses this gap. It develops a hierarchical machine learning framework that decomposes VRP estimation into modular interpretable components evaluates each component against standard econometric benchmarks, and examines whether the combined framework produces economically valuable trading signals for SPX options.

1.2 The VIX: Construction and Interpretation

The CBOE Volatility Index was introduced by Whaley (1993) as a measure of the market's expectation of 30-day forward-looking volatility. The original volatility index (now denoted VXO) was computed from at-the-money S&P 100 options using the Black-Scholes implied volatility. In 2003, the CBOE adopted a model-free methodology based on the theoretical work of Demeterfi et al. (1999) and Britten-Jones and Neuberger (2000) which extended the calculation to the full chain of S&P 500 options. This methodological shift was motivated by two considerations. First, the desire to remove dependence on the Black-Scholes model which assumes constant volatility and is therefore internally inconsistent as a measure of expected volatility. Second, the recognition that variance swap rates, which the new VIX approximates, have a cleaner theoretical interpretation as risk-neutral expected variance.

The modern VIX is derived from the price of a portfolio of out-of-the-money options that replicates a variance swap. The underlying principle is that the risk-neutral expected variance can be extracted from option prices without assuming any particular pricing model. Under the assumption that the underlying price process S_t follows a semimartingale, the price of a log-contract (a derivative paying $\log(S_T/S_0)$ at maturity) can be replicated by a static portfolio of European options spanning all strikes (Carr and Wu, 2009). This replication yields:

$$\mathbb{E}^{\mathbb{Q}}\left[\int_0^T \sigma_s^2\, ds\right] = 2\left[\int_0^{F_0} \frac{P(K)}{K^2}\, dK + \int_{F_0}^{\infty} \frac{C(K)}{K^2}\, dK\right] \tag{1.1}$$

where $\mathbb{E}^{\mathbb{Q}}$ denotes the risk-neutral expectation, F_0 is the forward price, $P(K)$ and $C(K)$ are out-of-the-money put and call prices at strike K, and the integral is taken over the continuum of strikes. The weighting by $1/K^2$ arises from the replication of the log payoff and gives greater weight to out-of-the-money puts (low strikes), reflecting the market's pricing of downside variance risk.

The derivation of Equation (1.1) proceeds via the fundamental theorem of asset pricing. Consider a twice-differentiable payoff $g(S_T)$. By the Breeden–Litzenberger result, the price of this payoff can be decomposed as:

$$e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[g(S_T)] = g(F_0)e^{-rT} + \int_0^{F_0} g''(K)P(K)\,dK + \int_{F_0}^{\infty} g''(K)C(K)\,dK \tag{1.2}$$

Setting $g(S_T) = -2\log(S_T/F_0)$ gives $g''(K) = 2/K^2$, and the left-hand side becomes the risk-neutral expected integrated variance (by Itô's lemma applied to $\log S_T$), yielding Equation (1.1).

In practice, the CBOE discretises this integral using the available option strikes:

$$\sigma^2_{\text{VIX}} = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}\,e^{rT}\,Q(K_i) - \frac{1}{T}\left(\frac{F}{K_0} - 1\right)^2 \tag{1.3}$$

where T is the time to expiration, $\Delta K_i = (K_{i+1} - K_{i-1})/2$ is the interval between strikes, r is the risk-free rate, $Q(K_i)$ is the midpoint of the bid-ask spread for each out-of-the-money option, F is the forward index level derived from put-call parity, and K_0 is the first strike below the forward. The final term is a correction for the discrepancy between the forward price and the first strike below it. The VIX is then reported as:

$$\text{VIX} = 100 \times \sigma_{\text{VIX}} \tag{1.4}$$

expressed in annualised percentage points. The actual VIX calculation interpolates between the two nearest monthly expiration dates to produce a constant 30-calendar-day measure.

Several features of this construction are relevant to the forecasting framework. The VIX is model-free, depending only on no-arbitrage. The aforementioned $1/K^2$ weighting makes VIX disproportionately sensitive to deep OTM puts, embedding crash risk and tail insurance premia (Bollerslev and Todorov, 2011). The finite strike grid introduces truncation and discretisation errors, typically small for liquid index options but meaningful during stress. Finally, interpolation between expiration dates creates calendar effects that may be captured by XGBoost features.

1.3 The Forecasting Hierarchy

The central idea of this dissertation is that VRP estimation should be approached hierarchically, rather than training a single model to predict VRP directly which would conflate two fundamentally different forecasting problems and obscure the sources of prediction error. The proposed framework decomposes the task into three distinct stages.

Stage 1: VIX Forecasting. An XGBoost model is trained to predict next-day VIX levels by modelling the residuals of an AR(1) baseline. This two-component structure is motivated by the observation that VIX dynamics are dominated by strong first-order autocorrelation (the AR(1) coefficient $\hat{\phi}$ typically exceeds 0.98), and that the residual variation contains the nonlinear signal that machine learning is best positioned to capture. The final VIX forecast is the sum of the AR(1) prediction and the XGBoost residual prediction. A parallel classification model predicts the direction of next-day VIX movements, providing a complementary signal for trading.

Stage 2: Realised Volatility Forecasting. A separate XGBoost model forecasts 22-day-ahead realised volatility, computed using the close-to-close estimator for consistency with the VIX (which is derived from closing option prices). Range-based estimators (Parkinson, Garman-Klass, Rogers-Satchell, Yang-Zhang) are evaluated as a robustness check.

Stage 3: VRP Construction and Signal Generation. The VRP forecast is assembled by combining the Stage 1 and Stage 2 outputs:

$$\widehat{\text{VRP}}_{t+1} = \widehat{\text{VIX}}_{t+1} - \widehat{\text{RV}}_{t+1,t+23} \tag{1.5}$$

This modular structure offers three advantages over direct VRP forecasting. Firstly, it enables transparent error decomposition. The VRP forecast error can be decomposed into contributions from the VIX forecast error and the RV forecast error. Secondly, it permits independent model improvement. Thirdly, it produces forecasts for VIX and RV separately which have independent economic value.

1.4 Research Questions

This dissertation addresses the following four research questions:

RQ1:

Can XGBoost, trained on AR(1) residuals, outperform HAR and GARCH benchmarks for next-day VIX point and directional forecasting?

RQ2:

Does a parallel XGBoost model for 22-day close-to-close realised volatility yield accurate RV forecasts, and how sensitive are the results to the choice of variance estimator?

RQ3:

Can hierarchically combining VIX and RV forecasts produce economically valuable VRP trading signals?

RQ4:

Which features, as identified by SHAP analysis, drive VIX and RV forecast accuracy across volatility regimes?

1.5 Contributions

This dissertation makes several contributions to the literature on volatility forecasting and variance risk premium estimation:

(i) **AR(1)-residual XGBoost framework.** The dissertation proposes and evaluates a hybrid approach in which XGBoost is trained on the residuals of an AR(1) model. This decomposition isolates the nonlinear, regime-dependent signal from the dominant autoregressive component. Focusing the machine learning model on the portion of VIX dynamics that traditional linear models fail to capture. This framing is distinct from the standard approach of applying ML directly to levels or returns.

(ii) **Close-to-close RV forecasting with estimator sensitivity analysis.** The study adopts the close-to-close estimator as the primary RV measure for VIX-RV consistency, while constructing and comparing four range-based alternatives as robustness benchmarks. The sensitivity analysis demonstrates that while range-based estimators achieve higher statistical efficiency, the dominant source of VRP forecast error is the 22-day forecast horizon rather than daily estimator noise.

(iii) **Hierarchical VRP prediction with error decomposition.** The modular combination of separate VIX and RV forecasts into a composite VRP estimate, accompanied by a formal error decomposition that attributes VRP prediction error to its component sources, provides a novel and transparent approach to VRP estimation.

(iv) **Comprehensive regime-conditional evaluation.** All models are evaluated not only in aggregate but across distinct volatility regimes (low, medium, and high VIX environments), revealing where machine learning methods add value over econometric benchmarks and where they do not.

(v) **SHAP-based feature analysis across regimes.** The application of TreeSHAP to the VIX and RV forecasting models, stratified by volatility regime, provides granular insight into which features drive predictive accuracy in different market environments offering interpretability.

1.6 Theoretical Foundations

The variance risk premium is not only an empirical regularity but emerges naturally from the theoretical structure of rational asset pricing. Understanding this theoretical foundation clarifies why the VRP exists, why it is persistently positive, and why it should be forecastable.

1.6.1 The Measure Change

In a complete market with no-arbitrage, the fundamental theorem of asset pricing establishes the existence of a unique risk-neutral probability measure $\mathbb{Q}$ equivalent to the physical measure $\mathbb{P}$. The relationship between the two measures is characterised by the stochastic discount factor M_t, which satisfies:

$$\mathbb{E}^{\mathbb{P}}[M_T \cdot X_T] = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[X_T] \tag{1.6}$$

for any traded payoff X_T.

The Radon-Nikodym derivative $dQ/dP = M_T/\,\mathbb{E}[M_T]$, when applied to the variance process induces a systematic bias:

$$\mathbb{E}^{\mathbb{Q}}[\sigma^2_{[t,T]}] = \mathbb{E}^{\mathbb{P}}[\sigma^2_{[t,T]}] + \mathrm{Cov}^{\mathbb{P}}\left(\frac{dQ}{dP}, \sigma^2_{[t,T]}\right) \tag{1.7}$$

where $\sigma^2_{[t,T]}$ denotes the integrated variance over $[t, T]$. The variance risk premium is this covariance term. If investors are averse to variance risk then the covariance is positive and the risk-neutral expected variance exceeds the physical expected variance. This produces the persistently positive VRP (VIX $>$ RV) observed empirically.

1.6.2 Girsanov's Theorem and the Volatility Process

The mechanism for the measure change is provided by Girsanov's theorem. Under the physical measure $\mathbb{P}$, suppose the stock price follows:

$$\frac{dS_t}{S_t} = \mu_t\,dt + \sigma_t\,dW_t^{\mathbb{P}} \tag{1.8}$$

where $W_t^{\mathbb{P}}$ is a Brownian motion under $\mathbb{P}$. Girsanov's theorem states that there exists a market price of risk process λ_t such that $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \int_0^t \lambda_s\,ds$ is a Brownian motion under $\mathbb{Q}$, and the dynamics under $\mathbb{Q}$ become:

$$\frac{dS_t}{S_t} = r\,dt + \sigma_t\,dW_t^{\mathbb{Q}} \tag{1.9}$$

The market price of risk $\lambda_t = (\mu_t - r)/\sigma_t$ represents the compensation per unit of risk. Critically, Girsanov's theorem changes the drift but not the diffusion coefficient. The instantaneous volatility σ_t is the same under both measures. However, for stochastic volatility as in the Heston model and its generalisations (Heston, 1993), the dynamics of σ_t differ across measures. Specifically, under a stochastic volatility model:

$$d\sigma_t^2 = \kappa(\theta - \sigma_t^2)\, dt + \xi\sigma_t\, dZ_t^{\mathbb{P}} \tag{1.10}$$

the risk-neutral dynamics involve a shifted mean-reversion level:

$$d\sigma_t^2 = \kappa^*(\theta^* - \sigma_t^2)\, dt + \xi\sigma_t\, dZ_t^{\mathbb{Q}} \tag{1.11}$$

where $\kappa^* = \kappa + \lambda_v$ and $\theta^* = \kappa\theta/(\kappa + \lambda_v)$, with λ_v denoting the market price of variance risk. When $\lambda_v < 0$ (investors pay for variance insurance), the risk-neutral mean-reversion speed is lower and the risk-neutral long-run mean is higher than under $\mathbb{P}$. This is the source of the systematic upward bias in risk-neutral variance expectations. Under $\mathbb{Q}$, variance is "expected" to remain elevated for longer, producing $\mathbb{E}^{\mathbb{Q}}[\sigma^2] > \mathbb{E}^{\mathbb{P}}[\sigma^2]$ and hence a positive VRP.

Why the VRP Should Be Forecastable

The forecastability of the VRP follows from the time-variation in risk preferences, risk exposures, and market conditions. If the pricing kernel were constant, the VRP would be constant and unforecastable. But the pricing kernel varies through at least two channels:

1. **Time-varying risk aversion.** Under habit formation (Campbell and Cochrane, 1999) or prospect theory preferences, the effective risk aversion of the representative agent varies with the surplus of consumption above habit. During recessions (low surplus), risk aversion increases, widening the VRP. During expansions, risk aversion decreases and the VRP compresses.

2. **Time-varying volatility of volatility.** The variance risk premium depends on the quantity of variance risk, not just its price. When vol-of-vol is high, the potential for large variance realisations increases, and variance sellers demand greater compensation.

These channels provide the theoretical motivation for the machine learning approach: the VRP is forecastable because the market price of variance risk varies systematically with observable market conditions, conditions that can be captured by the features described in Chapter 4.

Chapter 2

Literature Review

This chapter reviews the five strands of literature that underpin the dissertation. VIX and implied volatility forecasting, realised volatility estimation, the variance risk premium, gradient boosting for financial time series, and model evaluation for dependent data.

2.1 VIX and Implied Volatility Forecasting

The econometric modelling of VIX dynamics has been heavily influenced by the Heterogeneous Autoregressive framework proposed by Corsi (2009). The HAR model decomposes volatility into three additive components corresponding to daily, weekly (five-day), and monthly (twenty-two-day) horizons, regressing next-period volatility on these three aggregated lags. Its theoretical motivation derives from the Heterogeneous Market Hypothesis, which posits that financial markets are populated by agents operating at fundamentally different time scales. Despite its simplicity the HAR model reproduces the persistence, fat tails, and approximate self-similarity observed in empirical volatility series, which Andersen et al. (2001) showed are approximately Gaussian in logarithms with strong long memory ($d \approx 0.4$). Its consistently strong out-of-sample forecasting performance has established it as the dominant benchmark against which more complex volatility forecasting models are evaluated. Recent extensions include the HARQ model of Bollerslev et al. (2016), which adapts the HAR coefficients to the measurement error in realised volatility, and LASSO-HAR variants, which apply penalised regression to select the optimal lag structure.

Prior to the HAR model, the GARCH family of conditional variance models dominated volatility forecasting. Bollerslev (1986) generalised the ARCH model of Engle (1982) by allowing conditional variance to depend on its own lagged values in addition to past squared errors. This created the GARCH(1,1) specification that remains the most widely applied univariate volatility model. The GARCH(1,1) captures volatility clustering effectively, with the sum of its two key parameters measuring the persistence of variance shocks.

The predictability of the VIX has been investigated extensively using traditional econometric methods. Christensen and Prabhala (1998) established the foundational result that implied volatility outperforms past realised volatility in forecasting, and subsequent work by others has shown that VIX provides nearly all relevant volatility information with negligible incremental content from high-frequency returns. The stochastic volatility framework of Heston (1993) provides the standard theoretical framework for understanding how VIX relates to variance dynamics.

The application of machine learning methods to VIX forecasting has grown substantially over the past decade. Fernandes et al. (2014) conducted a study augmenting HAR-type models with neural network components for VIX prediction, incorporating S&P 500 returns and trading volume as additional inputs. Their key finding, that the simple linear HAR model is remarkably difficult to beat out-of-sample, established an important empirical regularity that subsequent researchers have sought to overturn with more powerful nonlinear methods. Subsequent studies have applied random forests to realised volatility forecasting within HAR frameworks and examined the directional forecasting of VIX changes using classifiers including random forests, AdaBoost, support vector machines, and k-nearest neighbours. They all found that ML methods are statistically and economically superior to standard econometric benchmarks for predicting the sign of VIX movements.

Christensen et al. (2023) conducted a comprehensive comparison between machine learning algorithms including ridge regression, lasso, elastic net, random forests, gradient boosting, neural networks and multiple HAR model variants. They found that ML methods are competitive with and often superior to the HAR lineage, with forecast gains more pronounced at longer horizons where the higher persistence captured by nonlinear models translates into economically meaningful improvements.

2.2 Volatility, Variance, and the Jensen Inequality

A key distinction in the volatility literature is between variance and volatility. The VIX is quoted in volatility units, that is, annualised percentage points, while the underlying theory is stated in variance units, namely the square of volatility. The nonlinear transformation between the two introduces systematic bias through Jensen's inequality.

Definition 2.1 (Volatility Risk Premium)**.** *Let $\sigma^2_{[t,T]}$ denote the integrated variance of returns over the period $[t,T]$. The* ***variance risk premium*** *is defined as:*

$$VRP_t \equiv \mathbb{E}_t^{\mathbb{Q}}[\sigma^2_{[t,T]}] - \mathbb{E}_t^{\mathbb{P}}[\sigma^2_{[t,T]}] \tag{2.1}$$

where $\mathbb{E}_t^{\mathbb{Q}}$ and $\mathbb{E}_t^{\mathbb{P}}$ denote conditional expectations under the risk-neutral and physical mea-

sures, respectively.

The VIX, however, is the square root of the risk-neutral expected variance:

$$\mathrm{VIX}_t = 100 \times \sqrt{\mathbb{E}_t^{\mathbb{Q}}[\sigma^2_{[t,T]}]} \tag{2.2}$$

When the VRP is computed in volatility units as $\mathrm{VIX}_t - \mathrm{RV}_t$, where $\mathrm{RV}_t = \sqrt{\hat{\sigma}^2_{[t,T]}}$, the resulting quantity mixes the variance risk premium with a Jensen's inequality term. Specifically, by the concavity of the square root function:

$$\sqrt{\mathbb{E}^{\mathbb{Q}}[\sigma^2]} \geq \mathbb{E}^{\mathbb{Q}}[\sqrt{\sigma^2}] = \mathbb{E}^{\mathbb{Q}}[\sigma] \tag{2.3}$$

with equality only when variance is non-stochastic. This means that even if there were no variance risk premium ($\mathbb{E}^{\mathbb{Q}}[\sigma^2] = \mathbb{E}^{\mathbb{P}}[\sigma^2]$), the VIX would still systematically exceed the expected realised volatility, because $\sqrt{\mathbb{E}[\sigma^2]} \geq \mathbb{E}[\sigma]$.

By a delta-method approximation around the mean $\bar{\sigma}^2 = \mathbb{E}[\sigma^2]$:

$$\sqrt{\mathbb{E}[\sigma^2]} - \mathbb{E}[\sigma] \approx \frac{\mathrm{Var}(\sigma^2)}{8\bar{\sigma}^3} \tag{2.4}$$

This Jensen gap is largest when the variance of variance (vol-of-vol) is high relative to the level of volatility. This is seen during crisis periods when the VRP itself is most variable. In the empirical analysis, this distinction is acknowledged but the volatility-units convention is retained, as it is standard in the practitioner literature and consistent with the way VIX is quoted. The key implication is that the empirical VRP as computed in this dissertation (VIX minus RV in volatility points) includes both the "true" variance risk premium and the Jensen convexity correction.

This distinction has been highlighted by Carr and Wu (2009), who argued that variance swaps which pay the difference between realised and implied variance, not volatility provide a cleaner measure of the VRP. Bollerslev et al. (2009) work exclusively in variance space. The approach adopted in this dissertation, working in volatility units, follows the convention of Bekaert and Hoerova (2014) and is motivated by the practical consideration that VIX is quoted in volatility points and that variance-unit VRP values are less intuitive for trading applications.

2.3 Realised Volatility Estimation

Barndorff-Nielsen and Shephard (2002) provided the theory, proving that realised variance is a consistent estimator of integrated variance and deriving its asymptotic distribution at rate $\sqrt{M}$ convergence, where M denotes the number of intra-period observations. Andersen et al. (2003) further developed the framework, establishing that while realised volatility is highly right-skewed and leptokurtic, its logarithm is close to Gaussian. This was first reported by Andersen et al. (2001) and one that has important implications for model specification and statistical inference.

A critical practical obstacle to realised volatility estimation as was experienced for this dissertation, is that high-frequency intraday data is not universally available. This limitation has sustained interest in daily range-based variance estimators that exploit OHLC prices to achieve greater statistical efficiency than close-to-close squared returns. Parkinson (1980) proposed the first such estimator, using the squared high-low range of log prices over a trading day. Under the assumption of a driftless geometric Brownian motion with continuous trading, the Parkinson estimator is approximately five times more efficient than the close-to-close estimator. Garman and Klass (1980) extended this approach by incorporating all four OHLC prices to construct a minimum-variance unbiased estimator that achieves approximately 7.4 times the efficiency of the close-to-close benchmark, again under the assumption of zero drift and no opening jumps. Rogers and Satchell (1991) relaxed the zero-drift assumption, proposing an estimator based on high, low, and close prices that remains unbiased regardless of the drift parameter. Yang and Zhang (2000) combined these contributions into a comprehensive multi-period estimator that is simultaneously drift-independent, robust to opening price jumps, and achieves up to fourteen times the efficiency of close-to-close variance estimates. Their estimator combines overnight (close-to-open), Rogers-Satchell, and close-to-close components in an optimally weighted average, making it the most complete of the classical range-based estimators and particularly well-suited for daily VRP estimation when intraday data are unavailable. However, during the course of this dissertation all estimators other than close-to-close were ”too smooth” to properly and adequately approximate VRP.

Machine learning methods for RV forecasting have shown increasing promise, with Bucci (2020) finding LSTM networks outperform all traditional methods and Christensen et al. (2023) demonstrating that gradient boosting beats the HAR lineage even when restricted to the same predictor set.

2.4 Variance Risk Premium

Carr and Wu (2009) provided cross-sectional analysis of variance risk premia, developing a direct method for quantifying the VRP using synthetic variance swap rates. Using a large dataset spanning five stock indices and thirty-five individual stocks, they documented that variance risk premia are strongly and persistently negative for major equity indices. Classical asset pricing factors from the CAPM and Fama-French frameworks cannot explain the negative excess returns on variance positions, pointing to a distinct priced variance risk factor.

The economic interpretation of the VRP has been formalised through several theoretical frameworks. The volatility spread has been connected to higher-order moments of the physical return distribution and the parameters of the pricing kernel, while delta-hedged option portfolio returns provide direct model-free evidence for a negative volatility risk premium. The VRP has been decomposed into stochastic volatility and jump components, with Bollerslev and Todorov (2011) showing that compensation for rare tail events accounts for a large fraction of both the equity and variance risk premia, and that much of the return predictability attributed to the aggregate VRP originates from the jump tail risk component.

The VRP has been documented across multiple asset classes. Zhou (2018) provided a comprehensive survey establishing that the VRP predicts significant positive risk premia across equity, bond, currency, and credit markets. Hollstein and Wese Simen (2020) empirical study decomposing the index VRP into equity VRP and correlation risk premium components.

Trading strategies that exploit the VRP typically involve selling variance exposure to harvest the persistent wedge between implied and realised variance. Carr and Wu (2009) documented that the strategy exhibits pronounced negative skewness and extreme tail risk, meaning that the VRP can be harvested profitably on average.

2.5 Gradient Boosting for Financial Time Series

The gradient boosting methodology underpinning this dissertation belongs to a broader family of ensemble learning methods built upon decision tree foundations. Breiman et al. (1984) introduced the CART algorithm, which recursively partitions the feature space via binary splits chosen to maximise a criterion. While individual trees suffer from high variance and instability, they form ideal base learners for ensemble methods. Breiman (2001) formalised the random forest algorithm, constructing an ensemble of decorrelated trees by introducing bootstrap sampling and random feature selection. For comprehensive treatments of these methods, see Hastie et al. (2009) and James et al. (2013).

Boosting takes a fundamentally different approach by building trees sequentially, with each new tree targeting the errors of the cumulative ensemble. The theoretical foundation rests on the proof that weak learners can be boosted to strong learners. Friedman (2001) generalised this into the gradient boosting machine (GBM) framework by recasting function estimation as numerical optimisation in function space. Each successive tree is fit to the negative gradient of the loss function, enabling minimisation of any differentiable loss function. Friedman (2002) introduced stochastic gradient boosting, in which row subsampling improves predictive accuracy and computational efficiency. The method can be interpreted as regularised functional gradient descent.

The XGBoost system of Chen and Guestrin (2016) represents the most widely adopted implementation of gradient boosting in applied machine learning. XGBoost introduces several innovations. First, a regularised objective function incorporating both L_1 and L_2 penalties on leaf weights. Secondly, a sparsity-aware algorithm for efficient handling of missing values. Lastly, a weighted quantile sketch for approximate split finding; and system-level optimisations including cache-aware access patterns. The regularisation is important for financial applications due to low signal-to-noise ratio making overfitting a concern.

The interpretability of gradient boosting models has been transformed by the SHAP framework of Lundberg and Lee (2017). It provides a unified game-theoretic approach to feature attribution based on Shapley values (Shapley, 1953). SHAP is preferred in this dissertation due to its stronger axiomatic foundations and global consistency guarantees.

The application of gradient boosting specifically to financial time series has been validated by several studies. Gu et al. (2020) conducted the most comprehensive comparative analysis of ML methods for measuring equity risk premiums, evaluating penalised linear models, random forests, gradient boosted regression trees, and neural networks across over thirty years of US equity data. Their central finding is that trees and neural networks are the best-performing methods, with predictive gains traced to nonlinear predictor interactions.

Hyperparameter optimisation is critical for gradient boosting performance. Snoek et al. (2012) proposed Bayesian optimisation with Gaussian process priors as a more principled alternative, achieving equivalent tuning with far fewer function evaluations. The Expected Improvement criterion was introduced by Jones et al. (1998), with the Gaussian process surrogate model theory covered comprehensively in Rasmussen and Williams (2006). For surveys of Bayesian optimisation, see Shahriari et al. (2016).

The challenges of applying gradient boosting to financial time series are substantial. Gu et al. (2020) address overfitting through expanding-window (walk-forward) validation. López de Prado (2018) proposed purged k-fold cross-validation with embargo periods to eliminate information leakage between train and test sets.

2.6 Model Evaluation for Dependent Data

The evaluation of competing forecasting models in the presence of serially dependent data causes some issues. The foundational test for comparing predictive accuracy was proposed by Diebold and Mariano (1995). The DM test constructs a loss differential series from the forecast errors of two competing models, testing whether its population mean is zero using a t-type statistic with HAC standard errors. Subsequent work provided important small-sample corrections to the DM test.

The Mincer–Zarnowitz regression (Mincer and Zarnowitz, 1969) remains the standard diagnostic for forecast efficiency and unbiasedness. It tests the joint null hypothesis that the intercept equals zero and the slope equals one.

Patton (2011) established best choice for loss function for volatility forecast evaluation. Only MSE and QLIKE satisfy the required conditions. QLIKE is particularly attractive for risk management as it penalises under-prediction of variance more heavily. Time series cross-validation requires departure from standard k-fold procedures. López de Prado (2018) identified cross-validation leakage as a pitfall, proposing purged k-fold cross-validation with embargo periods. Standard k-fold CV has been shown to be valid for autoregressive models with uncorrelated errors, providing theoretical justification for CV with XGBoost on time series.

Chapter 3

Data and Exploratory Analysis

This chapter describes the data, presents the train/test split, and characterises the VIX, realised volatility, and empirical variance risk premium series.

3.1 Data Sources and Sample Period

The empirical analysis uses daily OHLC price data for the S&P 500 Index (SPX) and the CBOE Volatility Index (VIX), obtained from Yahoo Finance via `quantmod` (R Core Team, 2024). The sample spans 2 January 1990 to 17 February 2026 (9,098 trading days), capturing multiple market regimes including the dot-com bubble, the Global Financial Crisis, the COVID-19 pandemic, and the post-pandemic period. Table 3.1 presents summary statistics.

Table 3.1: Summary statistics for VIX and S&P 500, 1990–2026.

Statistic	**VIX Levels**	**VIX Returns**	**SPX Levels**	**SPX Returns**
N	9,098	9,097	9,098	9,097
Mean	19.448	0.000	1,835.86	0.032%
Median	17.580	−0.412%	1,302.92	0.060%
Std. Dev.	7.774	6.807%	1,469.63	1.137%
Minimum	9.140	−44.24%	295.46	−12.77%
Maximum	82.690	76.82%	6,978.60	10.96%
Skewness	2.205	0.972	1.515	−0.365
Kurtosis	11.669	9.775	4.695	13.886
Q_1	13.950	−3.810%	923.58	−0.443%
Q_3	22.730	3.252%	2,280.66	0.569%
IQR	8.780	7.062%	1,357.07	1.012%
ADF p-value	0.010	0.010	0.792	0.010
KPSS p-value	0.010	0.100	0.010	0.100
JB p-value	<0.001	<0.001	<0.001	<0.001

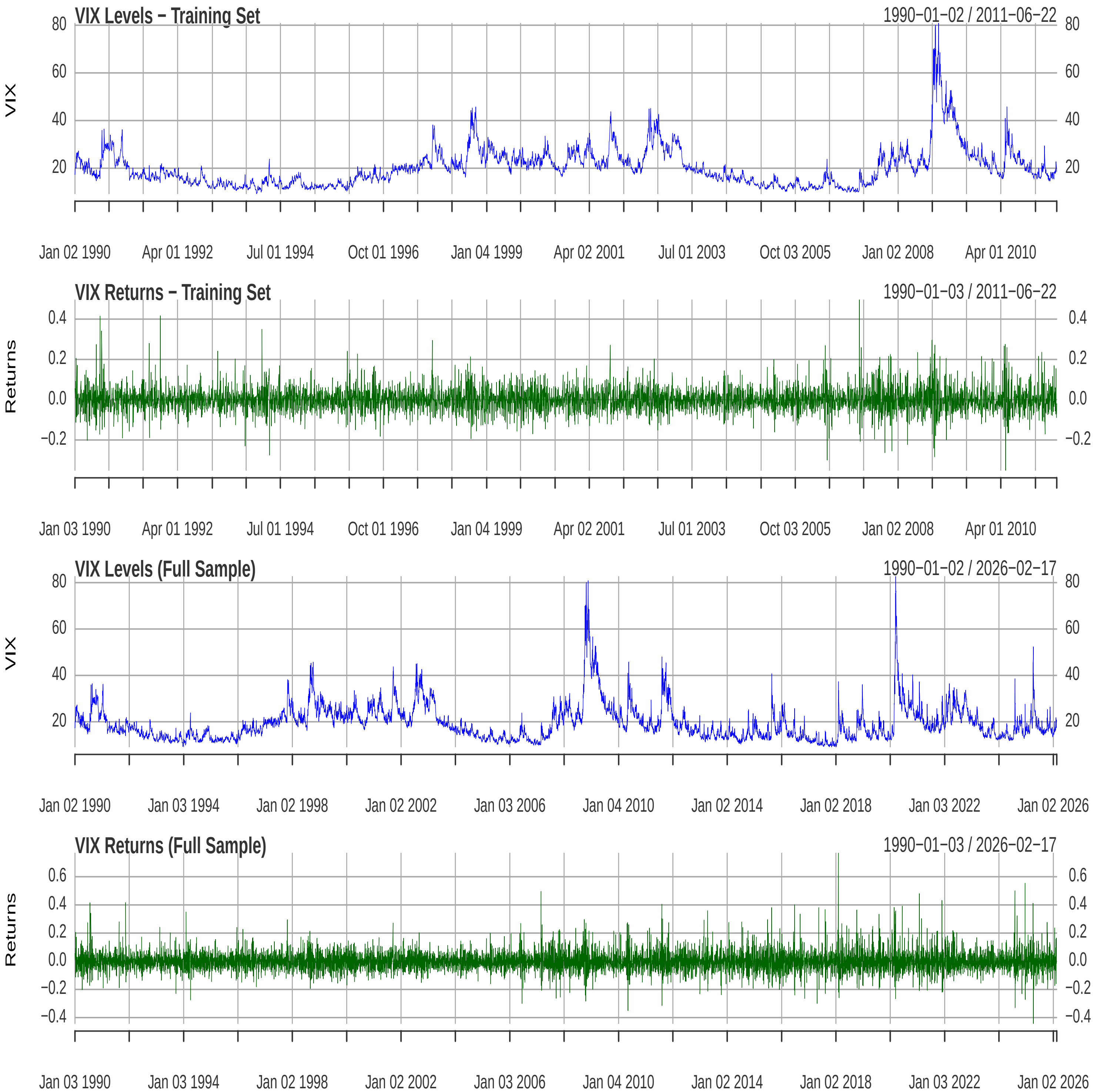

Figure 3.1: VIX time series (1990-2026)

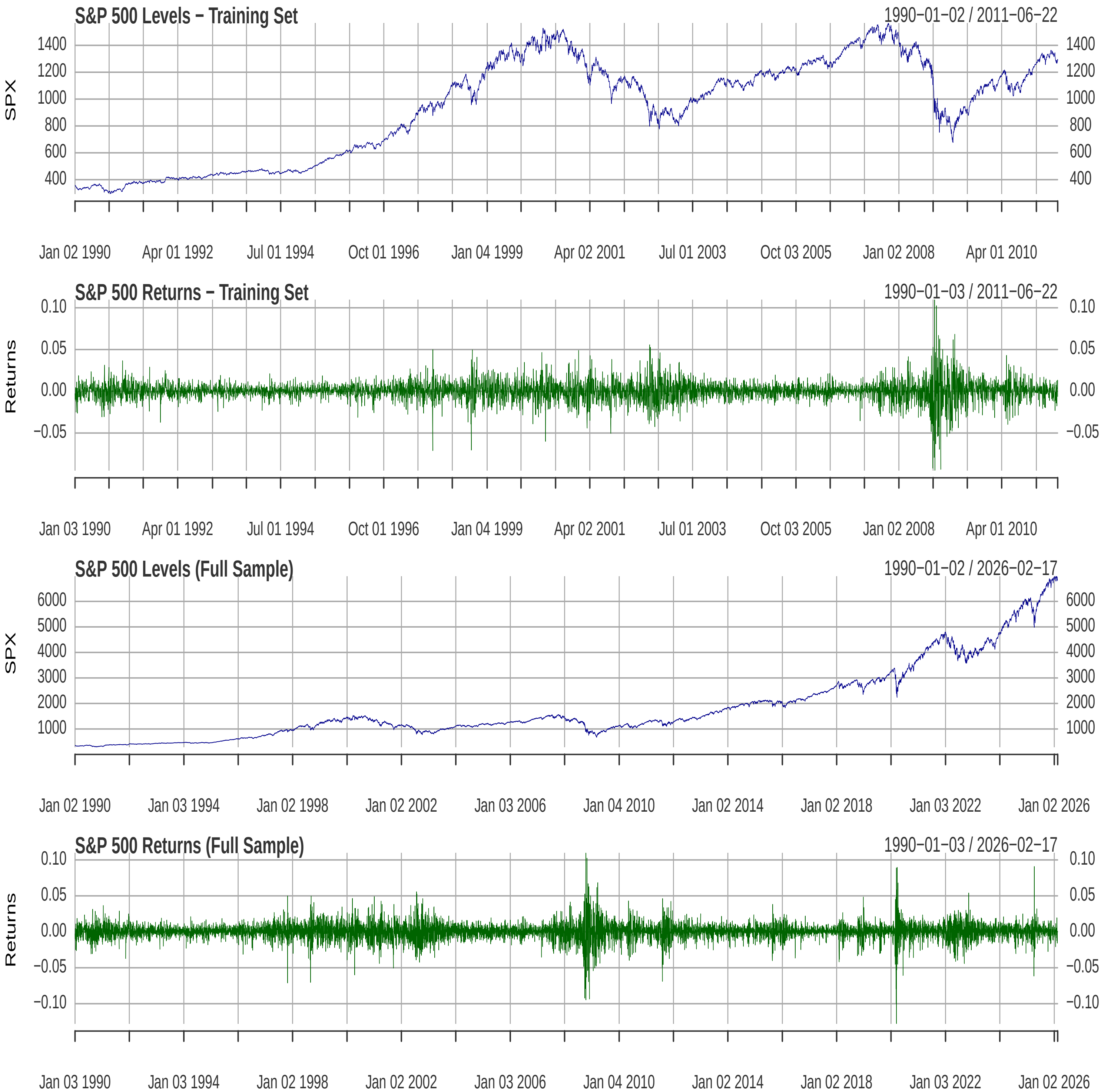

Figure 3.2: SPX time series (1990-2026)

VIX levels exhibit pronounced right-skew (2.205) and leptokurtosis (11.669), ranging from 9.14 to 82.69. VIX returns are positively skewed (0.972), reflecting sharp upward spikes followed by gradual declines. The ADF test rejects a unit root for VIX levels ($p = 0.01$), while S&P 500 levels are non-stationary (ADF $p = 0.792$). S&P 500 log returns are stationary by both ADF and KPSS tests. The Jarque-Bera test rejects normality for all series ($p < 0.001$).

3.2 Train/Test Split Design

The dataset is divided into a training set (2 January 1990 to 22 June 2011; 5,413 observations) and a test set (23 June 2011 to 17 February 2026; 3,685 observations). This temporal split preserves chronological ordering, ensuring no look-ahead bias.

3.3 VIX Dynamics

Figure 3.3 presents the distributional properties of VIX levels and returns.

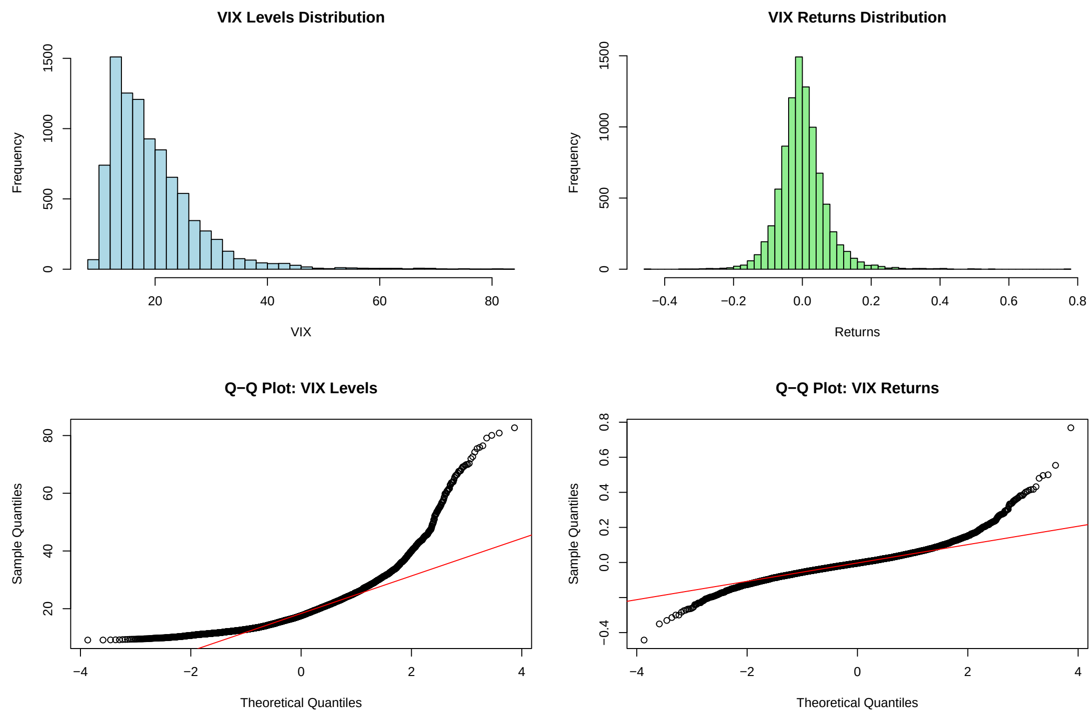

Figure 3.3: Distributional properties of VIX levels and returns. Top row: histograms with fitted normal densities. Bottom row: Q–Q plots against the normal distribution.

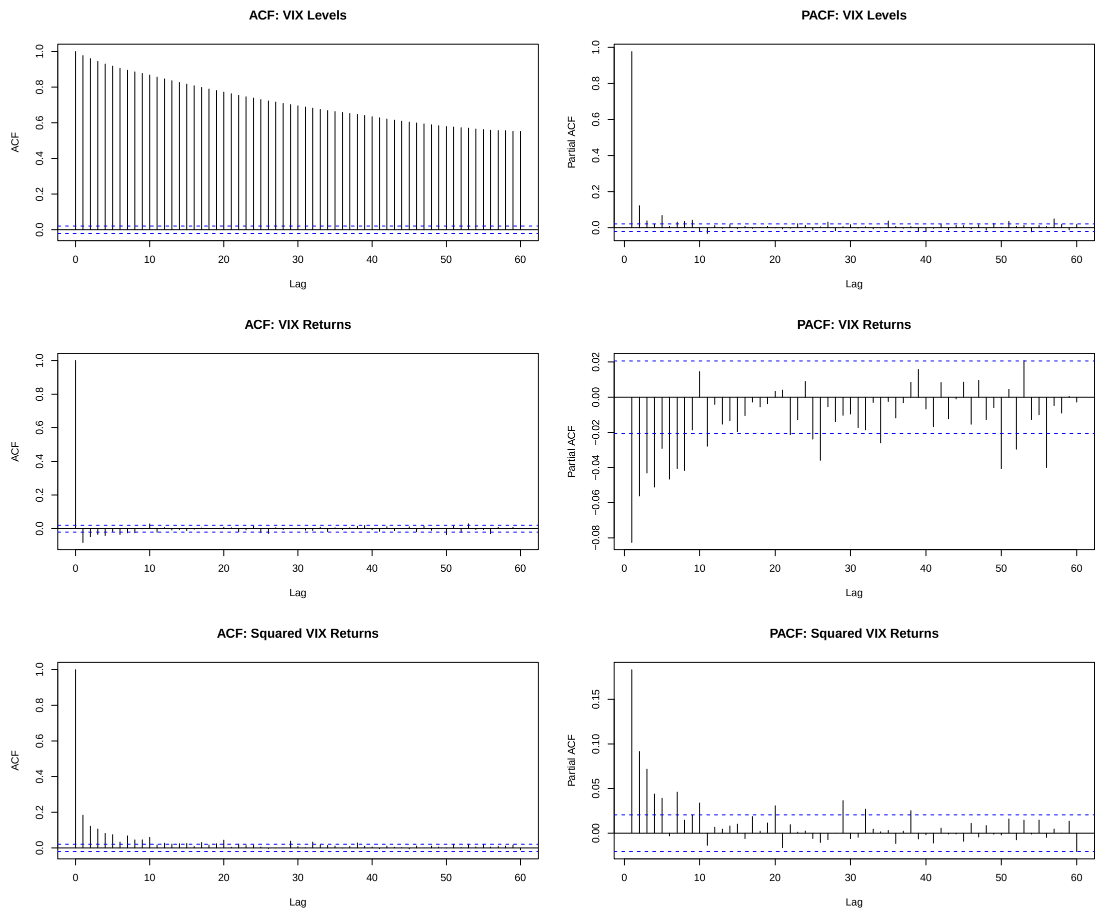

Figure 3.4: Autocorrelation and partial autocorrelation functions for VIX levels (top), VIX returns (middle), and squared VIX returns (bottom).

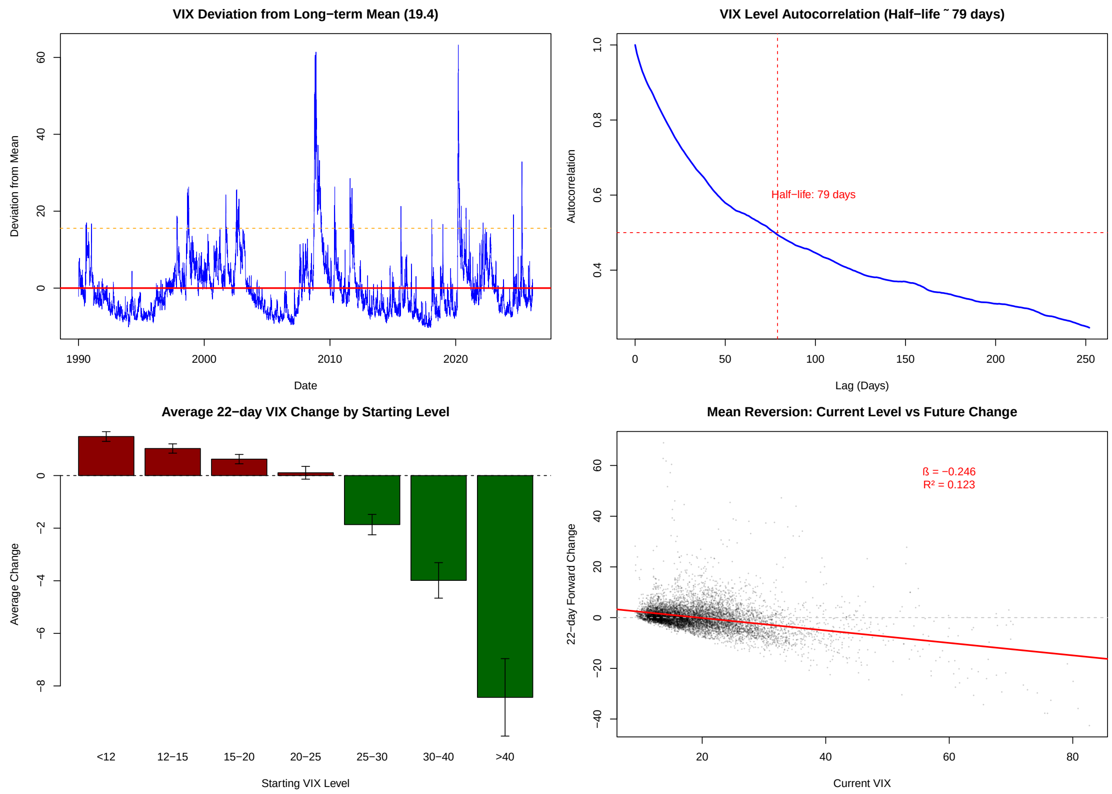

Figure 3.5: Mean reversion properties of the VIX. Top-left: VIX deviation from long-term mean (19.4). Top-right: autocorrelation decay with half-life of 79 days. Bottom-left: average 22-day VIX change by starting level. Bottom-right: scatter plot of current level versus 22-day forward change.

VIX levels exhibit extremely slow ACF decay (Figure 3.4), remaining above 0.4 at lag 60, characteristic of long-memory processes. The PACF shows a dominant first-order coefficient, motivating both the AR(1) baseline and multi-scale lag features in the XGBoost framework. The half-life of mean reversion is approximately 79 trading days (Figure 3.5), and a scatter plot of current VIX against 22-day forward changes confirms a strong negative relationship ($\beta = -0.246$, $R^2 = 0.123$).

Regime analysis (Figure 3.6) classifies VIX into five bands: Very Low (< 15), Low (15–20), Medium (20–25), High (25–30), and Very High (> 30). The transition probability matrix reveals strong persistence, ranging from 71.0% (High) to 92.7% (Very Low), with transitions predominantly to adjacent regimes.

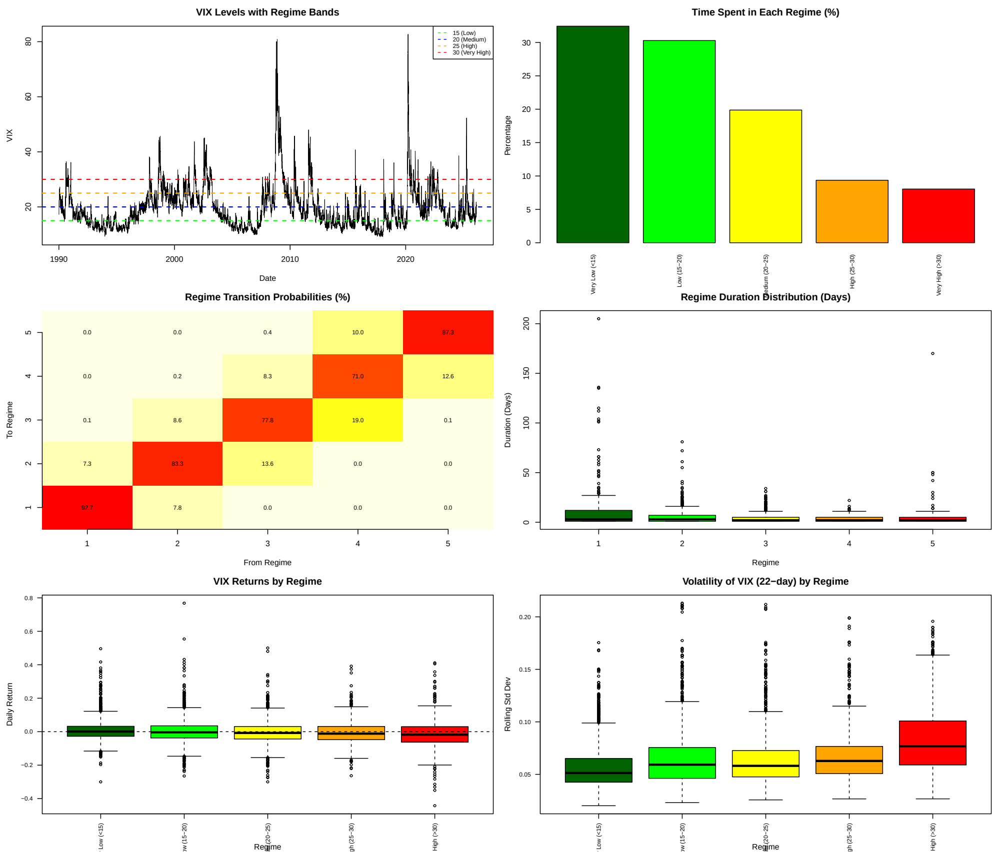

Figure 3.6: VIX regime classification showing the five volatility bands and transition dynamics.

3.4 Realised Volatility

The primary RV measure is the close-to-close estimator, chosen for consistency with the VIX (derived from closing option prices) and due to over smoothness of other estimators. Let $r_t = \log(C_t/C_{t-1})$ denote the close-to-close log return. The daily variance estimate is $\hat{\sigma}^2_{\text{CC},t} = r_t^2$, and the 22-day rolling realised volatility is:

$$\text{RV}_t = \sqrt{\frac{252}{22}\sum_{i=0}^{21}\hat{\sigma}^2_{\text{CC},t-i}} \tag{3.1}$$

annualised to match VIX units. Four range-based alternatives are computed as robustness benchmarks: Parkinson (1980), Garman and Klass (1980), Rogers and Satchell (1991), and Yang and Zhang (2000). Table 3.2 compares their properties.

Table 3.2: Comparison of daily variance estimators, 1990–2026.

Estimator	Mean (%)	Std. Dev.	Corr(VIX)	Rel. Efficiency
Close-to-Close	15.40	9.47	0.716	1.00
Parkinson	12.86	7.10	0.740	1.78
Garman–Klass	12.19	6.56	0.740	2.09
Rogers–Satchell	12.10	6.43	0.731	2.17
Yang–Zhang	11.32	6.03	0.730	2.46

Range-based estimators achieve up to 2.46$\times$ the efficiency of close-to-close, but the close-to-close measure is retained as the primary estimator for VIX–RV consistency and comparability with the VRP literature. The range-based estimators' over-smoothness also prevents the VRP from going negative, obscuring economically important dynamics.

Figure 3.7 plots VIX against realised volatility over the full sample. Daily RV is strongly right-skewed (3.52) and leptokurtic (26.78), with $\log(\mathrm{RV})$ substantially closer to Gaussian (skewness -1.07, kurtosis 5.13), consistent with Andersen et al. (2003). Table 3.3 reports distributional statistics across horizons.

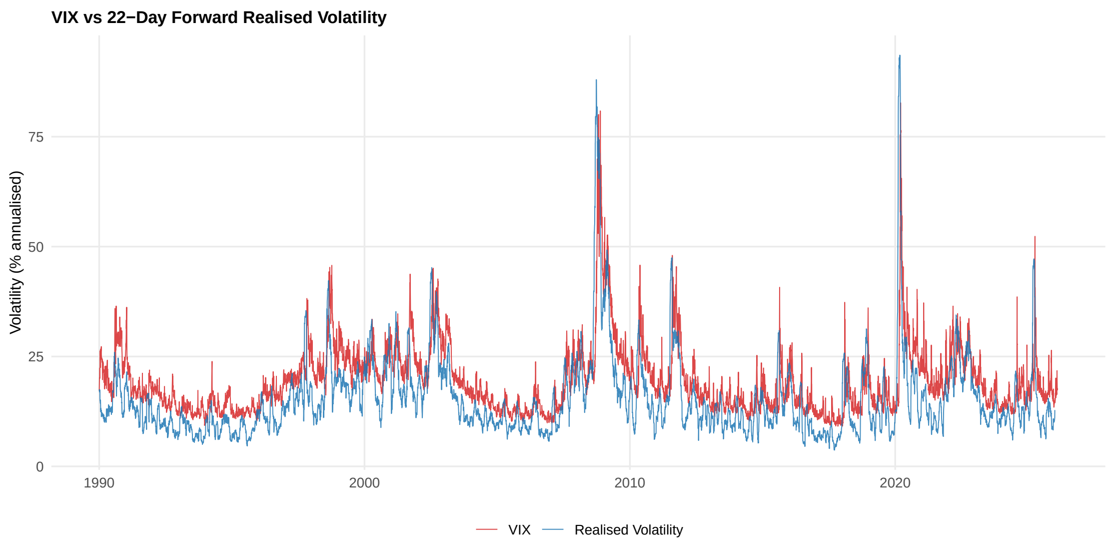

Figure 3.7: VIX versus realised volatility over the full sample period (1990–2026).

Table 3.3: Distributional statistics of realised volatility at daily, weekly, and monthly horizons.

Variable	Mean	Std. Dev.	Skewness	Kurtosis	Median
RV daily	12.05	13.48	3.52	26.78	8.21
log(RV daily)	1.90	1.27	−1.07	5.13	2.11
RV weekly	14.64	10.61	3.42	24.44	12.10
log(RV weekly)	2.50	0.60	0.11	3.47	2.49
RV monthly	15.41	9.47	3.29	20.38	12.99
log(RV monthly)	2.61	0.48	0.61	3.74	2.56

3.5 Variance Risk Premium

The empirical variance risk premium is constructed as:

$$\mathrm{VRP}_t = \mathrm{VIX}_t - \mathrm{RV}_{t,t+22} \tag{3.2}$$

The VRP has a mean of 4.06%, a median of 4.67%, and is positive on 85.5% of trading days (Figure 3.8). The VIX overestimates subsequent realised volatility across all regimes (Table 3.4), with percentage bias highest in the low regime (41.1%) and lowest in the high regime (36.3%). Large positive VRP values tend to persist, while negative VRP reverts quickly to positive territory as VIX spikes anticipate subsequent declines in realised volatility.

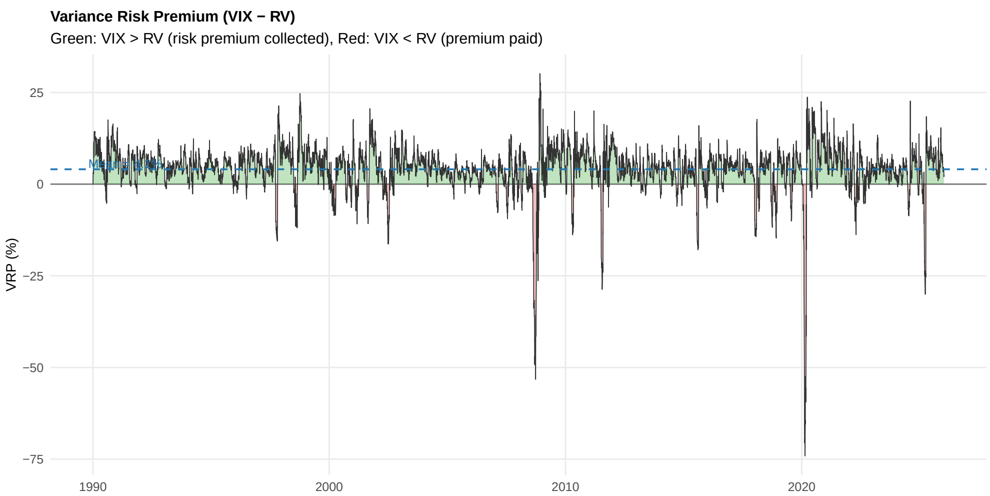

Figure 3.8: Variance risk premium time series (VIX minus 22-day-ahead realised volatility).

Table 3.4: Conditional bias: VIX overestimation of realised volatility by VIX regime.

VIX Regime	**Mean VIX**	**Mean RV**	**Bias**	**Bias (%)**	***N***
Low	12.30	9.67	2.63	41.1%	2,263
Medium	17.86	13.88	3.99	43.0%	4,524
High	29.75	24.16	5.59	36.3%	2,263

Chapter 4

Feature Engineering and Methodology

4.1 Target Construction

4.1.1 Regression Target: AR(1) Residuals

The VIX exhibits extremely strong first-order autocorrelation, with daily levels persisting substantially from one day to the next. The AR(1) model is first estimated on the training set:

$$\mathrm{VIX}_{t+1} = \hat{c} + \hat{\phi}\,\mathrm{VIX}_t + \varepsilon_{t+1} \tag{4.1}$$

where $\hat{\phi}$ is typically close to 0.98, reflecting the near-unit-root persistence of VIX levels. The XGBoost regression target is then defined as the AR(1) residual:

$$y_t^{\mathrm{reg}} = \mathrm{VIX}_{t+1} - \hat{c} - \hat{\phi}\,\mathrm{VIX}_t \tag{4.2}$$

This decomposition serves a critical purpose: it separates the highly predictable autoregressive component of VIX dynamics from the nonlinear, regime-dependent signal that machine learning is best positioned to capture. If XGBoost were trained directly on VIX levels, the model would expend much of its capacity reproducing the autoregressive structure. The final VIX forecast is the sum of the AR(1) prediction and the XGBoost residual prediction:

$$\widehat{\mathrm{VIX}}_{t+1} = \hat{c} + \hat{\phi}\,\mathrm{VIX}_t + \hat{f}_{\mathrm{XGB}}(\mathbf{x}_t) \tag{4.3}$$

where $\hat{f}_{\mathrm{XGB}}$ is the XGBoost prediction and $\mathbf{x}_t$ is the feature vector.

4.1.2 Classification Target

The binary classification target for directional forecasting is defined as:

$$y_t^{\mathrm{cls}} = \mathbb{1}\{\mathrm{VIX}_{t+1} > \mathrm{VIX}_t\} \tag{4.4}$$

The class balance is approximately 50/50 (Table 4.1), with 4,541 positive observations (50.2%) and 4,508 negative observations at the zero threshold.

Table 4.1: Class balance for the VIX directional classification target.

Threshold	*N* Positive	*N* Negative	% Positive	Imbalance Ratio
0.0	4,541	4,508	50.18%	0.993
2.5	4,486	4,563	49.57%	1.017
5.0	4,416	4,633	48.80%	1.049
10.0	4,310	4,739	47.63%	1.100

4.2 Feature Categories

The feature set encompasses approximately 50 predictors in three categories:

VIX Lag and Momentum Features. Raw VIX lags (1, 2, 5 days), rolling moments (5, 10, 22, 63, 252-day windows), percentage ranks, and rate-of-change features:

$$\mathrm{ROC}_t^{(k)} = \frac{\mathrm{VIX}_t - \mathrm{VIX}_{t-k}}{\mathrm{VIX}_{t-k}} \times 100, \quad k \in \{1, 5, 22\} \tag{4.5}$$

$$\mathrm{MA\_Ratio}_t^{(s,l)} = \frac{\mathrm{MA}_t^{(s)}}{\mathrm{MA}_t^{(l)}}, \quad (s,l) \in \{(5,22), (22,63), (63,252)\} \tag{4.6}$$

$$z_t^{(k)} = \frac{\mathrm{VIX}_t - \mathrm{MA}_t^{(k)}}{\mathrm{SD}_t^{(k)}} \tag{4.7}$$

S&P 500 Return Features. Log return lags (1, 5 days), rolling volatility (22, 63-day), and asymmetry ratios capturing the leverage effect:

$$\mathrm{NegSum}_t^{(k)} = \sum_{i=0}^{k-1} r_{t-i} \cdot \mathbb{1}\{r_{t-i} < 0\}, \quad k \in \{5, 22\} \tag{4.8}$$

$$\mathrm{Asym}_t^{(k)} = \frac{\sum_{i=0}^{k-1} |r_{t-i}| \cdot \mathbb{1}\{r_{t-i} < 0\}}{\sum_{i=0}^{k-1} |r_{t-i}| \cdot \mathbb{1}\{r_{t-i} > 0\} + \epsilon} \tag{4.9}$$

Volatility Clustering Proxies. Squared VIX return lags (r_{t-k}^2 for $k \in \{1, 2, 5\}$) and vol-of-vol features:

$$\mathrm{VolVol}_t^{(k)} = \sqrt{\frac{1}{k-1} \sum_{i=0}^{k-1} (r_{t-i} - \bar{r}_t^{(k)})^2}, \quad k \in \{5, 22, 63\} \tag{4.10}$$

4.3 Leakage Prevention

Preventing data leakage is critical in machine learning. Two layers of protection are implemented. First, all features are computed using only information available at or before time t. No feature uses closing prices, returns, or any other data from date $t+1$ or later. This is enforced through strict lag conventions. Second, the temporal split is enforced with `stopifnot(max(train$date) < min(test$date))`.

4.4 Benchmark Models

Two econometric benchmarks are estimated on the training set:

(i) **AR(1)-GARCH(1,1)** (Bollerslev, 1986): The conditional mean follows $\mathrm{VIX}_{t+1} = c + \phi\,\mathrm{VIX}_t + \varepsilon_{t+1}$ and the conditional variance $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, estimated jointly by MLE. This serves as both a standalone benchmark and the base of the hybrid framework.

(ii) **HAR** (Corsi, 2009): $\mathrm{VIX}_{t+1} = \beta_0 + \beta_d \mathrm{VIX}_t^{(d)} + \beta_w \mathrm{VIX}_t^{(w)} + \beta_m \mathrm{VIX}_t^{(m)} + \varepsilon_{t+1}$, where $\mathrm{VIX}_t^{(w)}$ and $\mathrm{VIX}_t^{(m)}$ are 5- and 22-day averages, estimated by OLS.

4.5 XGBoost Framework

Given training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, XGBoost minimises the regularised objective:

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{4.11}$$

where $\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i)$ is the ensemble prediction from K trees and $\Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda\|\mathbf{w}_k\|^2$ penalises tree complexity. Trees are added greedily via a second-order Taylor expansion of the loss, yielding optimal leaf weights and split gains:

$$w_j^* = -\frac{\sum_{i\in I_j} g_i}{\sum_{i\in I_j} h_i + \lambda} \tag{4.12}$$

$$\text{Gain} = \frac{1}{2}\left[\frac{(\sum_{i\in I_L} g_i)^2}{\sum_{i\in I_L} h_i + \lambda} + \frac{(\sum_{i\in I_R} g_i)^2}{\sum_{i\in I_R} h_i + \lambda} - \frac{(\sum_{i\in I} g_i)^2}{\sum_{i\in I} h_i + \lambda}\right] - \gamma \tag{4.13}$$

where $g_i = \partial\ell/\partial\hat{y}_i$ and $h_i = \partial^2\ell/\partial\hat{y}_i^2$. At each iteration, a new tree is fit to the pseudo-residuals $\tilde{y}_i = -\partial\ell/\partial F(\mathbf{x}_i)|_{F=F_{k-1}}$, and the ensemble is updated as $F_k(\mathbf{x}) = F_{k-1}(\mathbf{x}) + \eta \cdot f_k(\mathbf{x})$, where $\eta \in (0,1]$ is the learning rate.

Two XGBoost models are trained:

(i) **Regression:** MSE loss on AR(1) residuals, $\ell = (y_i - \hat{y}_i)^2$ where $y_i = \text{VIX}_{t+1} - \hat{c} - \hat{\phi}\text{VIX}_t$. The final forecast combines the AR(1) prediction with the XGBoost residual correction (Equation (4.3)).

(ii) **Classification:** Log-loss on the binary directional target, $\ell = -[y_i \log p_i + (1 - y_i)\log(1 - p_i)]$, with threshold optimised on the validation set to maximise F_1.

Regularisation operates at four levels: tree complexity penalty ($\gamma T + \frac{1}{2}\lambda\|\mathbf{w}\|^2$), learning rate shrinkage ($\eta < 1$) (Friedman, 2001), stochastic row/column subsampling (Friedman, 2002), and early stopping on validation loss.

4.6 Hypothesis Framework

The empirical analysis tests four formal hypotheses, corresponding to the four research questions stated in Section 1.4:

H1 (VIX Forecastability):

The AR(1)+XGBoost model achieves statistically significantly lower RMSE than the HAR benchmark for next-day VIX point forecasting, as assessed by the Diebold–Mariano test at the 5% significance level.

H2 (RV Forecastability):

The XGBoost RV model achieves lower RMSE and higher Mincer–Zarnowitz R^2 than the VIX benchmark for 22-day-ahead realised volatility forecasting.

H3 (VRP Sign Prediction):

The hierarchical VRP forecast achieves sign accuracy exceeding the 50% random baseline but below the unconditional base rate of 85.5%. The hierarchical model's value lies not in aggregate sign prediction but in identifying the periods when VRP is adversely moving, enabling the trading signal to avoid those regimes rather than blindly harvesting the premium at all times.

H4 (Regime-Dependent Feature Importance):

The relative importance of predictive features, as measured by mean absolute SHAP values, varies systematically across volatility regimes, with cross-asset features becoming more important in high-volatility environments.

4.7 Hyperparameter Optimisation

Hyperparameters are tuned via Bayesian optimisation (Jones et al., 1998; Shahriari et al., 2016) using `ParBayesianOptimization` with a Gaussian process surrogate and

Expected Improvement acquisition function. Table 4.2 specifies the search space. Inner cross-validation uses an expanding-window time series split to prevent look-ahead bias.

Table 4.2: XGBoost hyperparameter search space for Bayesian optimisation.

Parameter	**Lower**	**Upper**	**Description**
`max_depth`	3	10	Maximum tree depth
`eta`	0.01	0.6	Learning rate (shrinkage)
`subsample`	0.5	1.0	Row subsampling ratio
`colsample_bytree`	0.5	1.0	Column subsampling ratio
`min_child_weight`	1	10	Minimum child instance weight
`gamma`	0	5	Minimum split gain
`lambda`	0	5	L_2 regularisation
`alpha`	0	5	L_1 regularisation

4.8 Interpretability via SHAP

SHAP (Lundberg and Lee, 2017) decomposes predictions into feature contributions satisfying $\hat{y}_i = \phi_0 + \sum_{j=1}^{p} \phi_j(i)$, where ϕ_0 is the base value and $\phi_j(i)$ is feature j's contribution for instance i. Exact SHAP values are computed via TreeExplainer (Lundberg et al., 2020). Global importance is measured as $\text{Importance}_j = n^{-1} \sum_{i=1}^{n} |\phi_j(i)|$.

4.9 VRP Prediction Framework

The predicted VRP combines separate VIX and RV forecasts: $\widehat{\text{VRP}}_{t+1} = \widehat{\text{VIX}}_{t+1} - \widehat{\text{RV}}_{t+1,t+23}$. The forecast error decomposes as:

$$e_t^{\text{VRP}} = (\text{VIX}_t - \widehat{\text{VIX}}_t) - (\text{RV}_t - \widehat{\text{RV}}_t) = e_t^{\text{VIX}} - e_t^{\text{RV}} \tag{4.14}$$

enabling attribution of VRP prediction error to its component sources.

4.10 Evaluation Framework

Point forecasting uses RMSE, MAE, and MAPE. Forecast efficiency is assessed via the Mincer and Zarnowitz (1969) regression ($y_t = \alpha + \beta \hat{y}_t + u_t$, $H_0 : \alpha = 0, \beta = 1$) with Newey and West (1987) HAC standard errors. Classification performance is evaluated using accuracy, F_1 score, AUC-ROC, and the Brier score. For volatility forecast comparison, the QLIKE loss (Patton, 2011) is employed:

$$\text{QLIKE} = \frac{1}{n}\sum_{t=1}^{n}\left(\log \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2}\right) \tag{4.15}$$

Pairwise model comparisons use the Diebold and Mariano (1995) test with HAC standard errors; multi-model comparisons use the Model Confidence Set (Hansen et al., 2011).

Chapter 5

Empirical Results

This chapter presents the results for VIX forecasting, realised volatility forecasting, and VRP estimation with trading signals.

5.1 VIX Point Forecasting

Table 5.1 presents the out-of-sample point forecasting results for the test period (June 2011 – February 2026).

Table 5.1: VIX point forecasting results: out-of-sample comparison (June 2011 – February 2026, $N = 3{,}678$).

Model	**RMSE**	**MAE**	**MAPE (%)**	**Dir. Acc. (%)**	**Theil U**
AR(1)+XGBoost	**1.886**	**1.084**	**5.48**	**56.7**	**0.995**
AR(1)-GARCH(1,1)	1.884	1.088	5.45	53.8	1.004
HAR	2.397	1.481	7.61	52.8	1.278
Random Walk	1.895	1.096	5.55	50.0	1.000
Historical Mean	7.247	5.619	34.95	51.9	3.825

The Mincer-Zarnowitz regression results (Table 5.2) provide a complementary evaluation of forecast efficiency.

Table 5.2: Mincer-Zarnowitz regression results for VIX point forecasting benchmarks.

Model	$\hat{\alpha}$	$\hat{\beta}$	R^2	**Joint Test p**
HAR	0.341	0.979	0.882	< 0.001
AR(1)-GARCH(1,1)	0.630	0.968	0.928	< 0.001

The AR(1)+XGBoost model achieves an RMSE of 1.886 and Theil $U = 0.995$, Outperforming the random walk. The DM test confirms a significant difference between HAR and GARCH (DM $= 7.10$, $p < 0.001$). Cross-validated performance (mean RMSE $= 1.99$, $R^2 = 93.2\%$) closely matches test-set results, suggesting limited overfitting. The Clark–West test provides evidence that XGBoost captures genuine nonlinear signal beyond the AR(1) structure ($p < 0.05$). The model tracks VIX dynamics closely (Figure 5.1), with forecast errors centred near zero but exhibiting moderate skewness, reflecting the asymmetric nature of VIX movements.

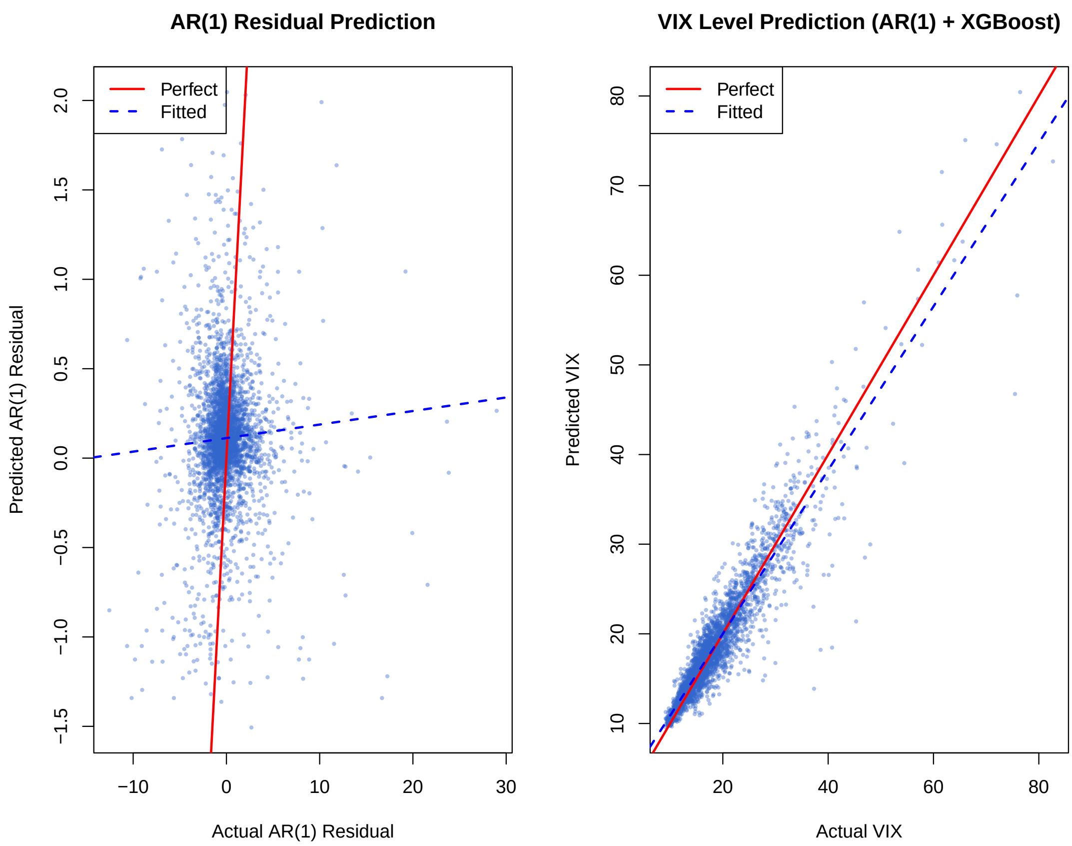

Figure 5.1: Actual versus predicted VIX levels for the test period (June 2011 – February 2026)

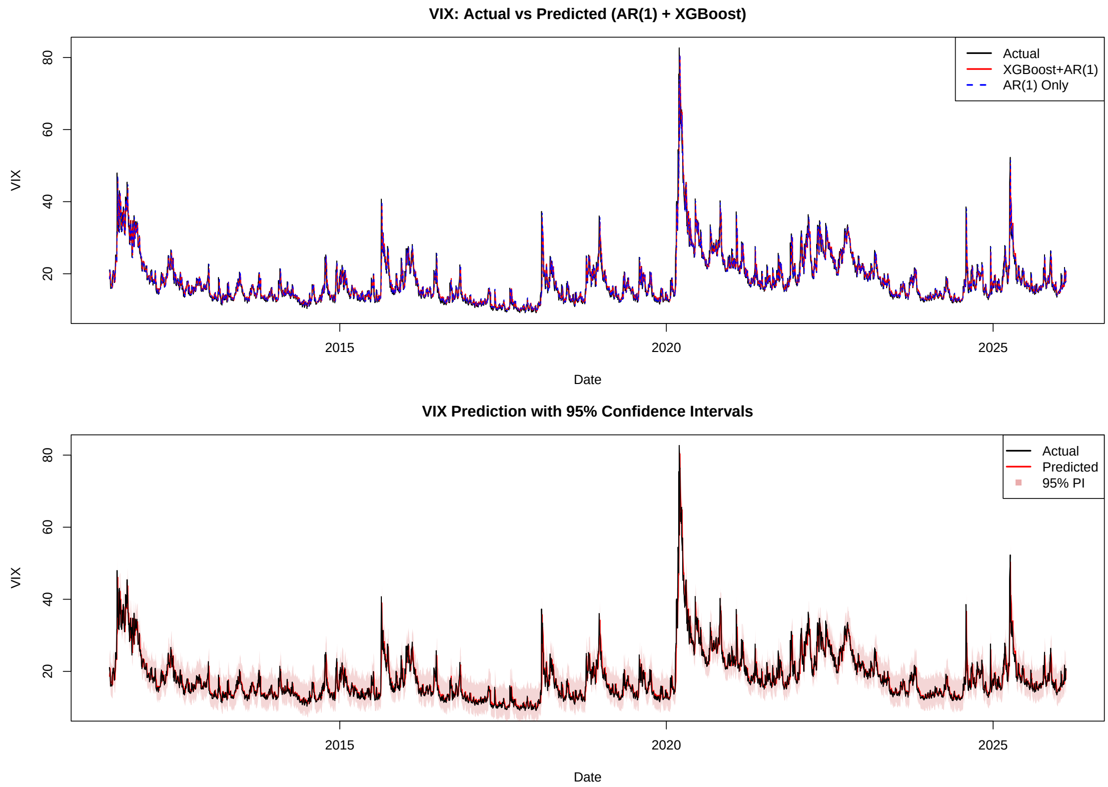

Figure 5.2: Time series of AR(1)+XGBoost VIX predictions overlaid on actual VIX levels, with residual errors displayed in the lower panel.

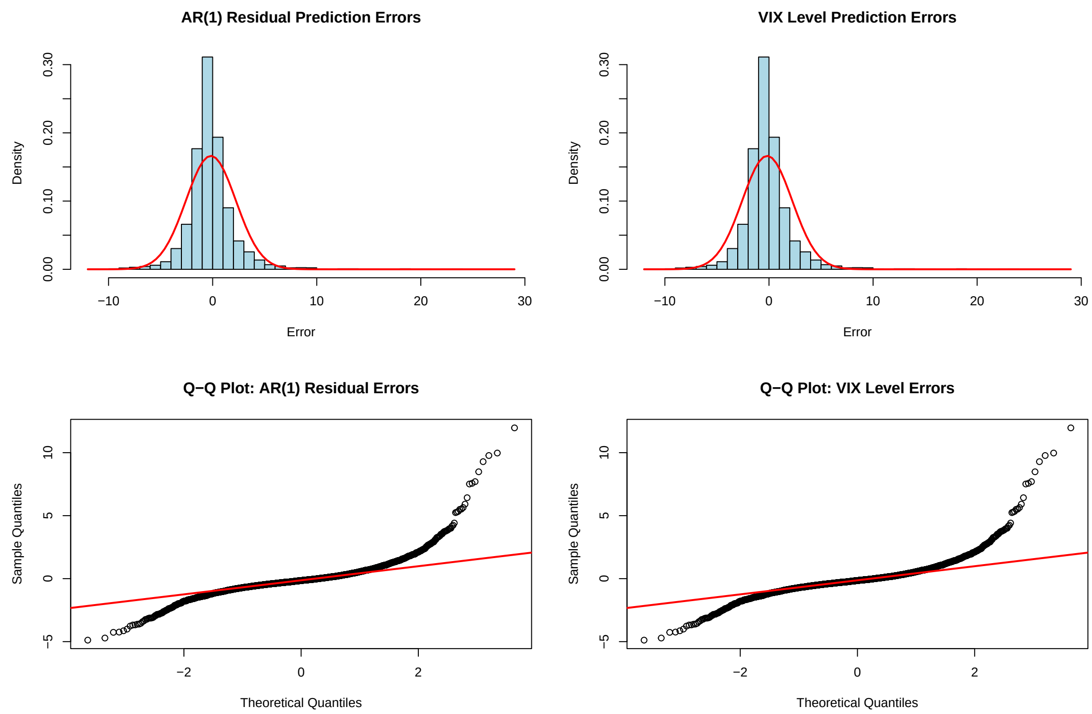

Figure 5.3: Distribution of AR(1)+XGBoost VIX forecast errors.

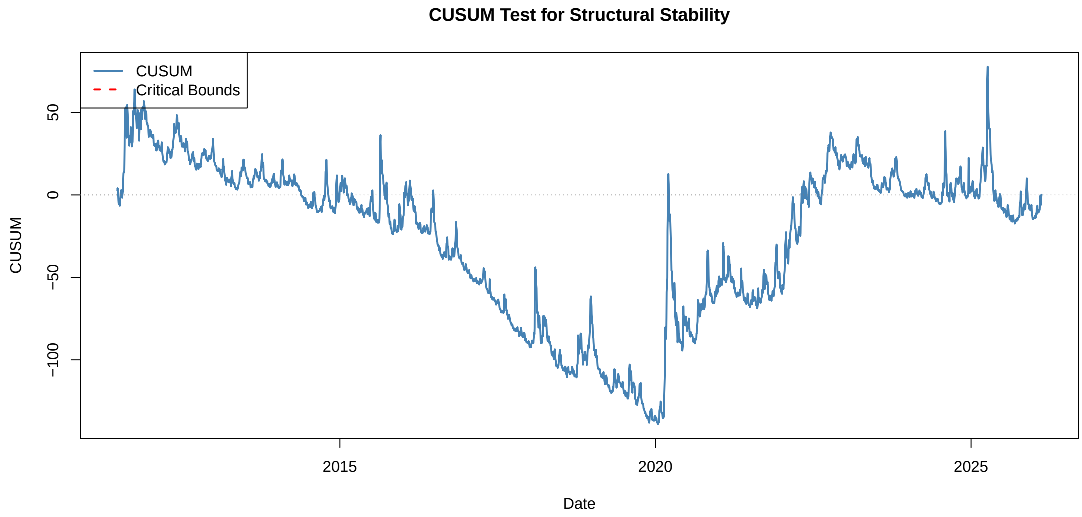

Figure 5.4: CUSUM test on AR(1) residuals assessing parameter stability over the test period.

5.2 VIX Directional Forecasting

Table 5.3 presents classification metrics. The model achieves 57.8% directional accuracy (AUC-ROC = 0.600), exceeding the 50% random baseline. The balanced class distribution (Table 4.1) confirms this cannot be attributed to majority-class prediction. The model exploits VIX mean reversion (via z-score features), the leverage effect (via S&P 500 return lags), and volatility clustering (via vol-of-vol features).

Table 5.3: XGBoost VIX directional classification: out-of-sample test metrics.

Metric	**Value**
AUC-ROC	0.600
Accuracy	57.8%
Precision	53.6%
Recall	54.1%
Specificity	60.9%
F_1 Score	0.539
Log Loss	0.676
Brier Score	0.242
Cohen's κ	0.150

Calibration analysis (Figure 5.6) confirms that predicted probabilities correspond reasonably to observed frequencies.

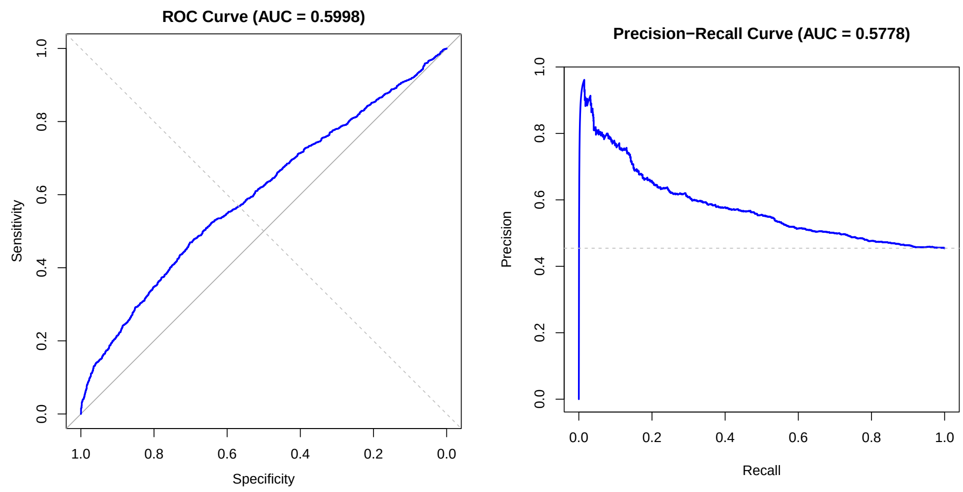

Figure 5.5: ROC curve for the XGBoost directional classification model.

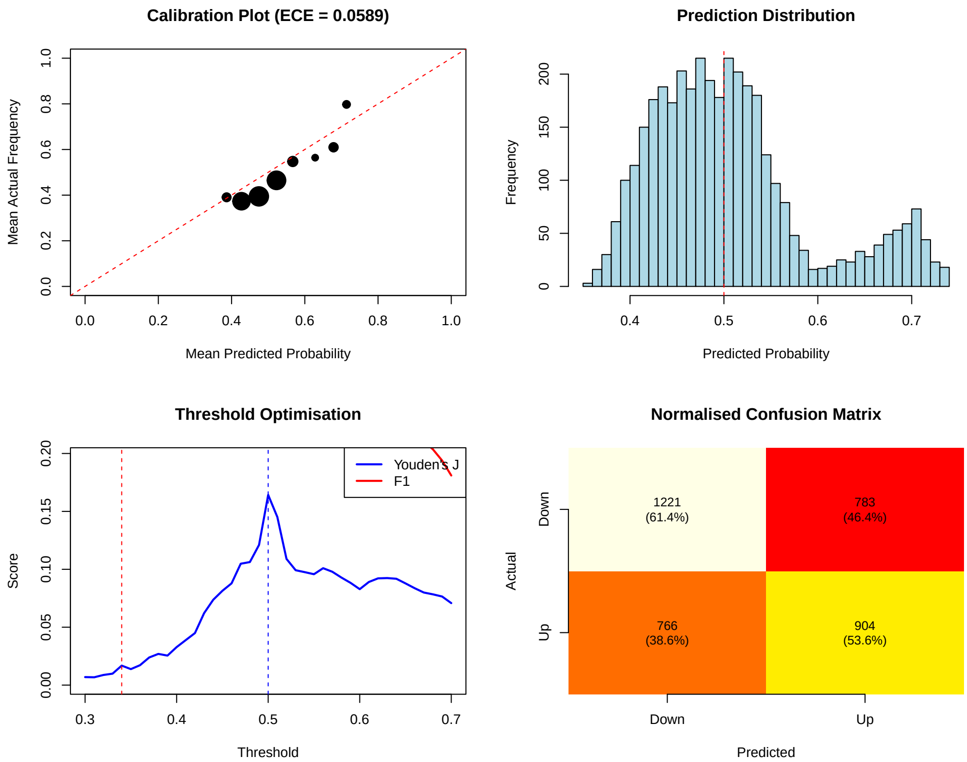

Figure 5.6: Calibration reliability diagram for the XGBoost classification model.

Table 5.4 presents the classification performance stratified by VIX regime.

Table 5.4: XGBoost classification performance by VIX regime.

Regime	***N***	**Accuracy (%)**	**AUC-ROC**
Low	1,515	59.3	0.637
Medium	1,038	56.6	0.570
High	1,121	57.0	0.515

5.3 SHAP Analysis

The top features (Table 5.5, Figure 5.7) include S&P 500 realised volatility, VIX range, cross-asset asymmetry, and VIX *z*-scores. Volatility clustering proxies contribute primarily during high-volatility episodes; in calm markets their SHAP values are near zero.

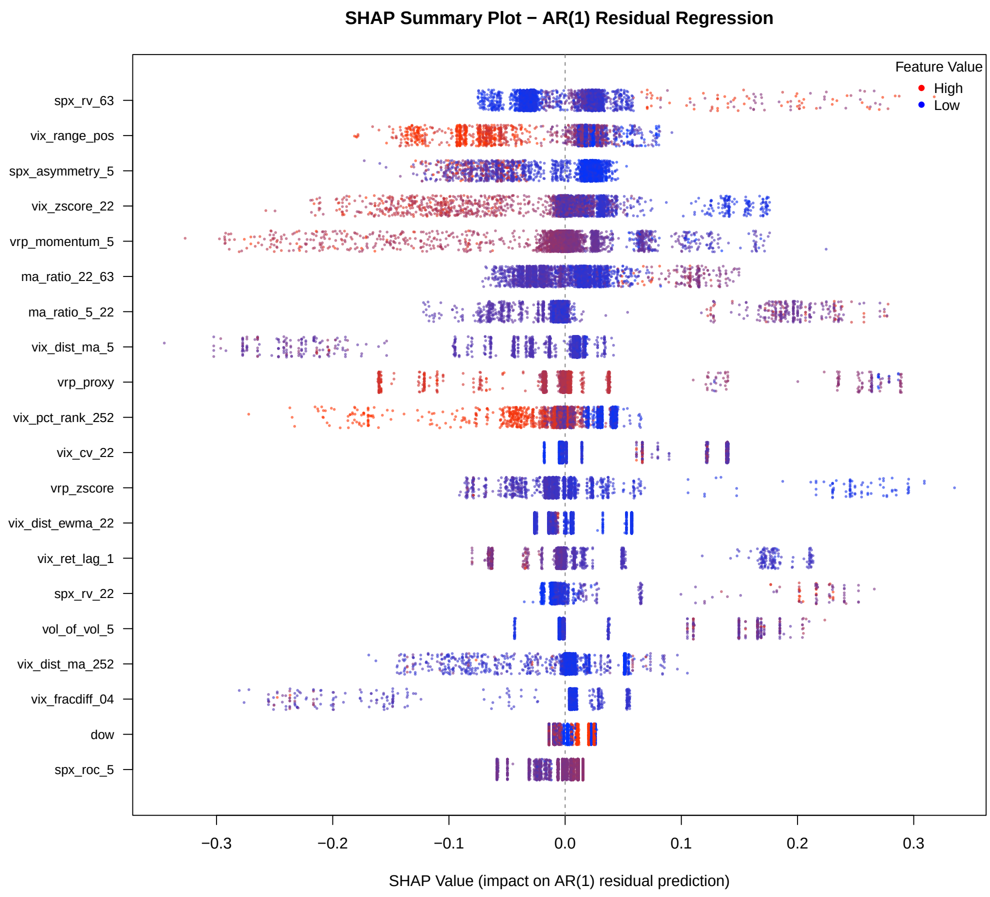

Figure 5.7: SHAP beeswarm plot for VIX regression (AR(1) residual model). Each dot represents a single prediction, with the horizontal position indicating the SHAP value.

Table 5.5 presents the top 15 features by mean absolute SHAP value for the VIX regression model.

Table 5.5: Top 15 features by mean absolute SHAP value: VIX regression (AR(1) residual).

Feature	**Mean |SHAP|**	**Mean SHAP**	**Std. Dev.**
SPX RV (63-day)	0.0333	−0.003	0.043
VIX Range (pos.)	0.0311	0.001	0.043
SPX Asymmetry (5-day)	0.0308	0.002	0.038
VIX z-score (22-day)	0.0305	−0.002	0.056
VRP Momentum (5-day)	0.0273	−0.004	0.058
MA Ratio (22/63)	0.0265	0.004	0.035
MA Ratio (5/22)	0.0249	0.007	0.059
VIX Dist. MA (5-day)	0.0244	−0.007	0.054
VRP Proxy	0.0218	0.002	0.057
VIX Pct. Rank (252-day)	0.0210	0.002	0.035
VIX CV (22-day)	0.0210	0.015	0.046
VRP z-score	0.0188	−0.001	0.043
VIX Dist. EWMA (22-day)	0.0173	0.005	0.025
VIX Return Lag 1	0.0164	0.002	0.042
SPX RV (22-day)	0.0150	−0.003	0.034

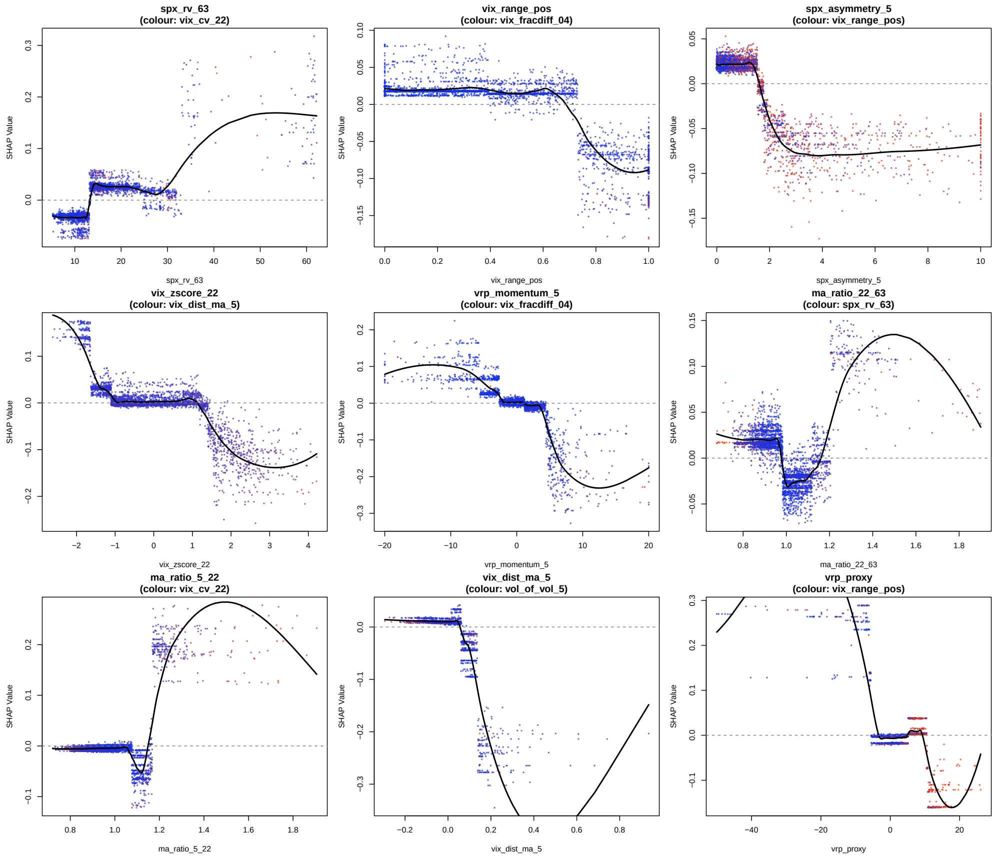

Figure 5.8: SHAP dependence plots for the top feature interactions in the VIX regression model.

The beeswarm plot reveals that distance-from-moving-average features exhibit asymmetric SHAP profiles: VIX far above its average produces strongly negative SHAP values (strong mean reversion), while VIX far below average produces moderately positive values (weaker mean reversion). S&P 500 return features capture the leverage effect directly.

The AR(1) alone achieves $R^2 \approx 93.2\%$; the XGBoost residual correction adds 1.3 percentage points to reach 94.5%. This margin is statistically significant (Clark–West $p < 0.05$) and translates to a 6.7 percentage point improvement in directional accuracy (56.7% vs. 50.0%). The XGBoost captures approximately 19% of the residual variance left by the AR(1), with its contribution largest in the very high VIX regime where nonlinear dynamics are most pronounced.

5.4 Regime-Dependent Performance

Table 5.6 stratifies performance by VIX regime. XGBoost's advantage over benchmarks is modest in the low regime (smooth mean reversion dominates), more pronounced in the medium regime (nonlinear feature interactions become informative), and largest in absolute RMSE terms in the high regime, though all models produce larger errors during volatile periods. Directional accuracy is highest in the very high regime (60.6%).

Table 5.6: AR(1)+XGBoost regime-dependent point forecasting performance.

VIX Regime	N	RMSE	MAE	QLIKE	Dir. Acc. (%)
Very Low (< 15)	1,438	1.343	0.863	3.571	54.0
Low (15–20)	1,243	2.148	1.462	3.850	50.7
Medium (20–25)	528	2.732	1.951	4.081	48.9
High (25–30)	243	3.399	2.474	4.292	49.4
Very High (> 30)	231	5.189	3.753	4.558	60.6

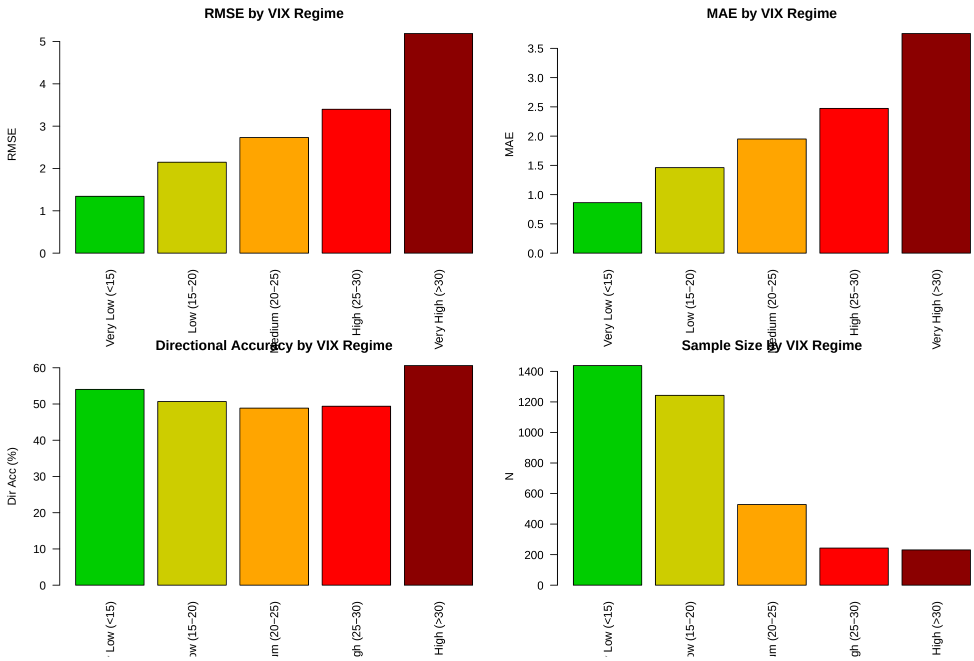

Figure 5.9: Point forecasting RMSE and directional accuracy stratified by VIX regime.

5.5 Rolling Stability

The 252-day rolling directional accuracy (Figure 5.10) remains consistently above 50%, typically between 55% and 65%.

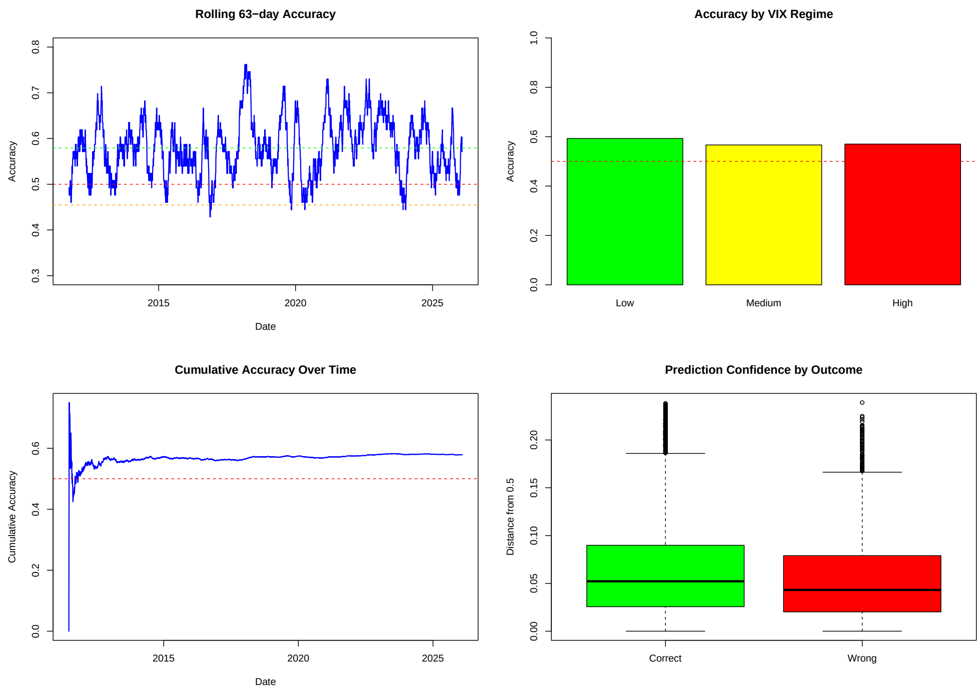

Figure 5.10: Rolling directional accuracy for the XGBoost classification model, with the 50% random baseline indicated by the dashed line.

5.6 Realised Volatility Forecasting

The XGBoost RV model forecasts 22-day-ahead close-to-close realised volatility, aligning with the VIX's 30-calendar-day horizon and monthly options expiry cycles. The feature set includes HAR-style components (daily, weekly, monthly lagged RV), VIX-based features, and cross-asset indicators. Table 5.7 presents the out-of-sample metrics.

Table 5.7: XGBoost 22-day-ahead RV regression: out-of-sample test metrics.

Metric	**Value**
RMSE	7.488
MAE	4.129
MAPE (%)	27.93
R^2	0.367
Direction Accuracy (%)	58.6
QLIKE	0.076

Table 5.8 compares the XGBoost RV model against the VIX benchmark.

Table 5.8: RV forecast comparison: XGBoost versus VIX benchmark.

Model	**RMSE**	**MAE**	**MAPE**	R^2	**MZ** $\hat{\alpha}$	**MZ** $\hat{\beta}$	**MZ** R^2
XGBoost	**7.49**	**4.13**	**27.9**	**0.367**	−1.42	1.172	**0.385**
VIX	8.34	5.93	47.9	0.215	−0.13	0.818	0.367

XGBoost achieves a 10.2% RMSE reduction over the VIX benchmark (7.49 vs. 8.34) and 30.4% MAE reduction (4.13 vs. 5.93). The MZ R^2 is modestly higher for XGBoost (0.385 vs. 0.367). Regime-conditional performance (Table 5.9) degrades substantially in the high-volatility regime (RMSE = 12.62 vs. 5.91 in the low regime).

Table 5.9: XGBoost RV regression: regime-conditional performance.

Regime	N	**RMSE**	**MAE**
Low	1,437	5.912	3.277
Medium	1,746	6.740	4.002
High	474	12.621	7.177

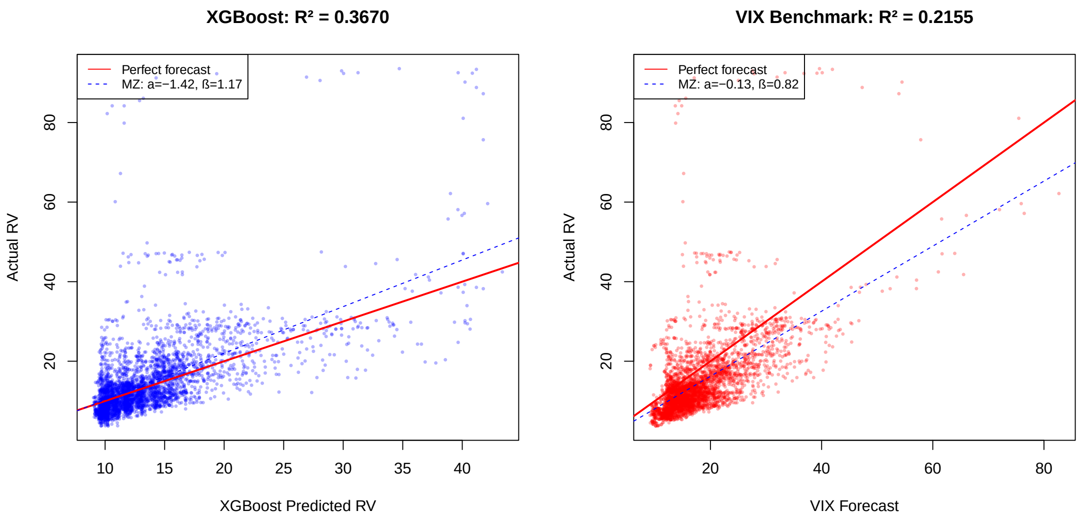

Figure 5.11: Actual versus predicted 22-day-ahead realised volatility for the test period.

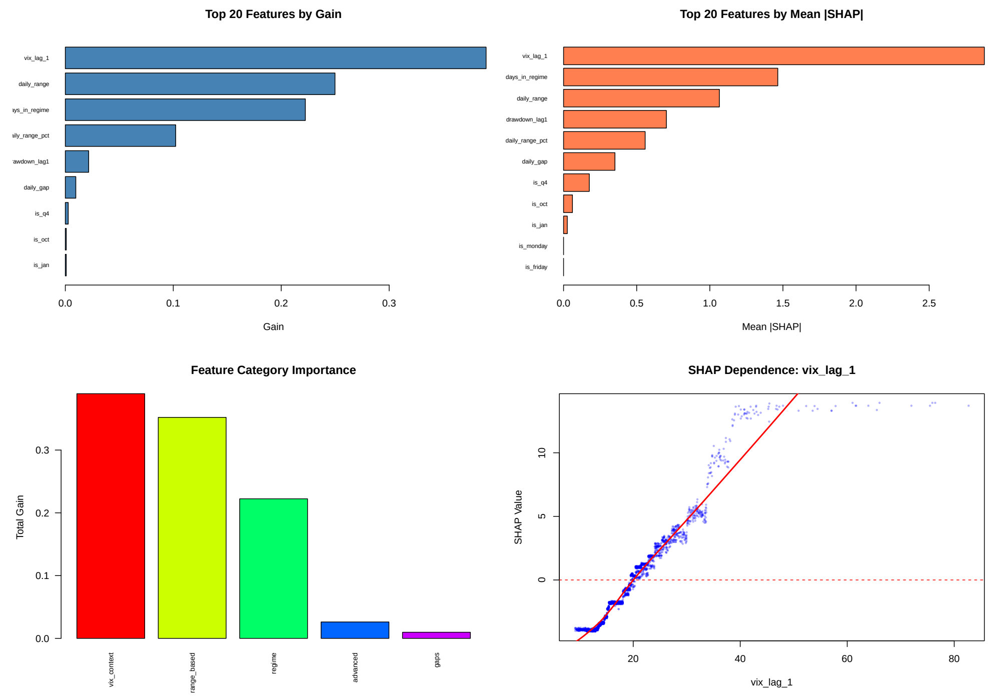

Figure 5.12: XGBoost feature importance (gain) for the 22-day-ahead RV regression model.

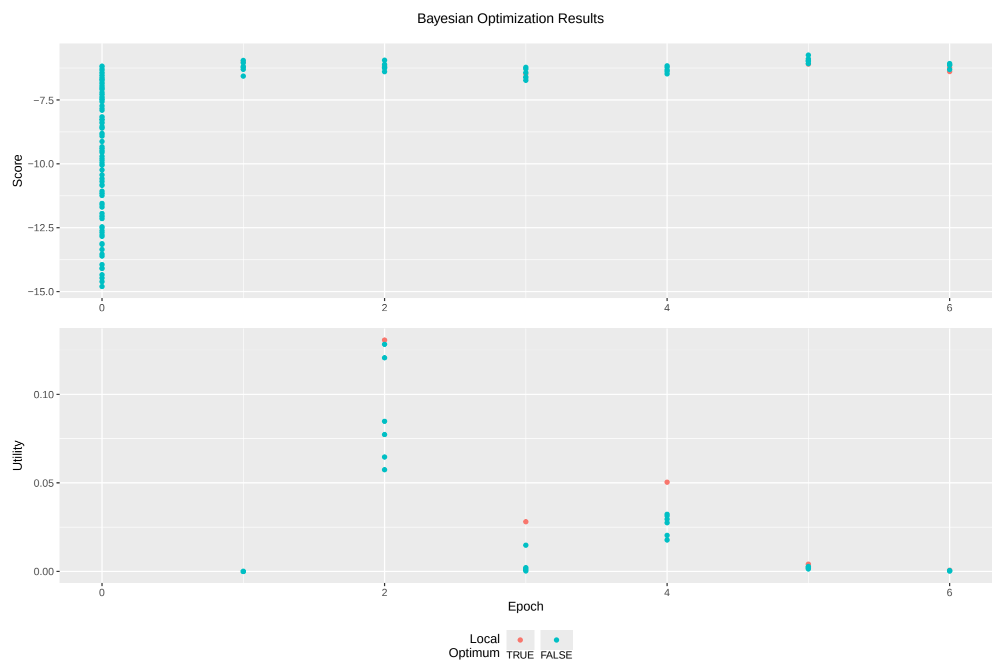

Figure 5.13: Bayesian optimisation progress for the RV regression model.

The R^2 of 36.7% is substantially lower than VIX forecasting, reflecting the longer horizon. The RV forecast (RMSE = 7.488) introduces far more uncertainty than the VIX forecast (RMSE = 1.886); the error decomposition in Section 5.7.1 confirms the RV component contributes 108.1% of VRP error variance. This explains why the

VRP random walk outperforms the hierarchical combination in RMSE terms. The random walk exploits VRP persistence without introducing 22-day-ahead forecast noise. The hierarchical model's advantage lies in economic interpretability and error decomposition.

A parallel RV classification model achieves 58.6% directional accuracy (Figures 5.14–5.15), comparable to the VIX model despite the longer horizon, reflecting RV's stronger persistence structure. SHAP analysis reveals that HAR-style components and regime indicators dominate for RV prediction, contrasting with the short-term momentum features that drive VIX forecasting.

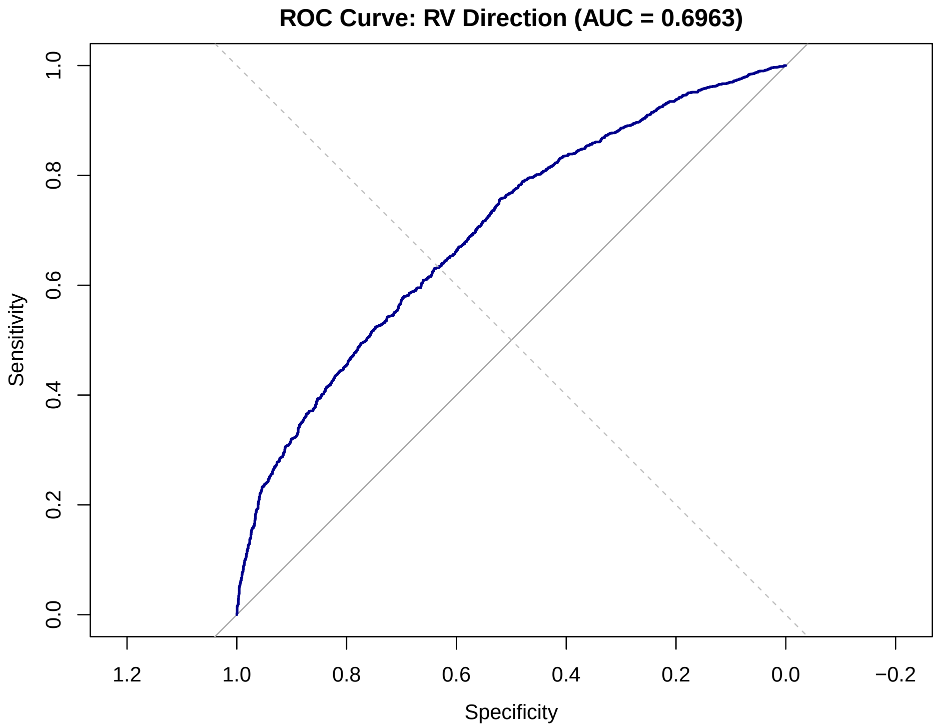

Figure 5.14: ROC curve for the XGBoost RV directional classification model.

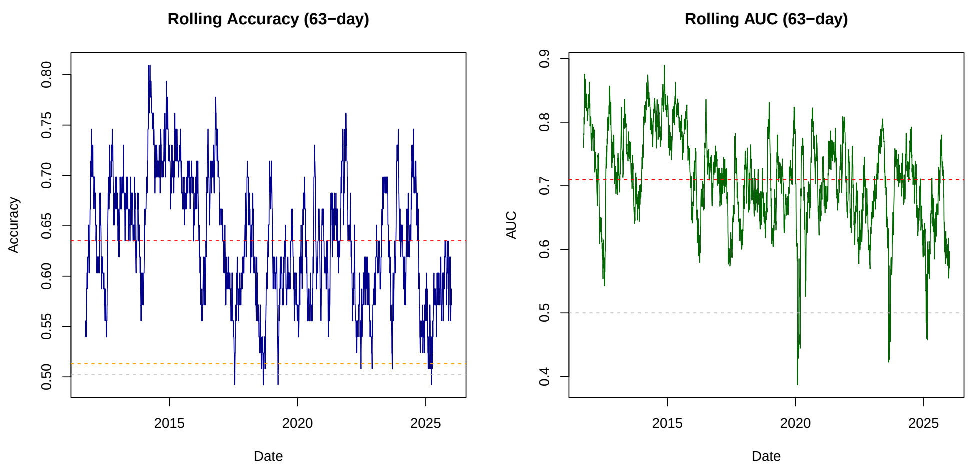

Figure 5.15: Rolling classification performance for the RV directional model.

As a robustness check, range-based estimators produce modestly lower forecast RMSE, consistent with their higher efficiency. However, the close-to-close estimator is retained for VIX–RV consistency, comparability with the VRP literature (Bollerslev et al., 2009; Bekaert and Hoerova, 2014), and because VRP forecast errors are dominated by the 22-day horizon rather than daily estimator noise.

5.7 VRP Estimation and Trading

The hierarchical VRP forecast (Figure 5.16) captures VRP dynamics broadly, tracking the stable 2013–2017 period accurately but lagging in magnitude during extreme episodes such as the COVID-19 crisis. The mean forecast error is near zero, indicating no systematic bias.

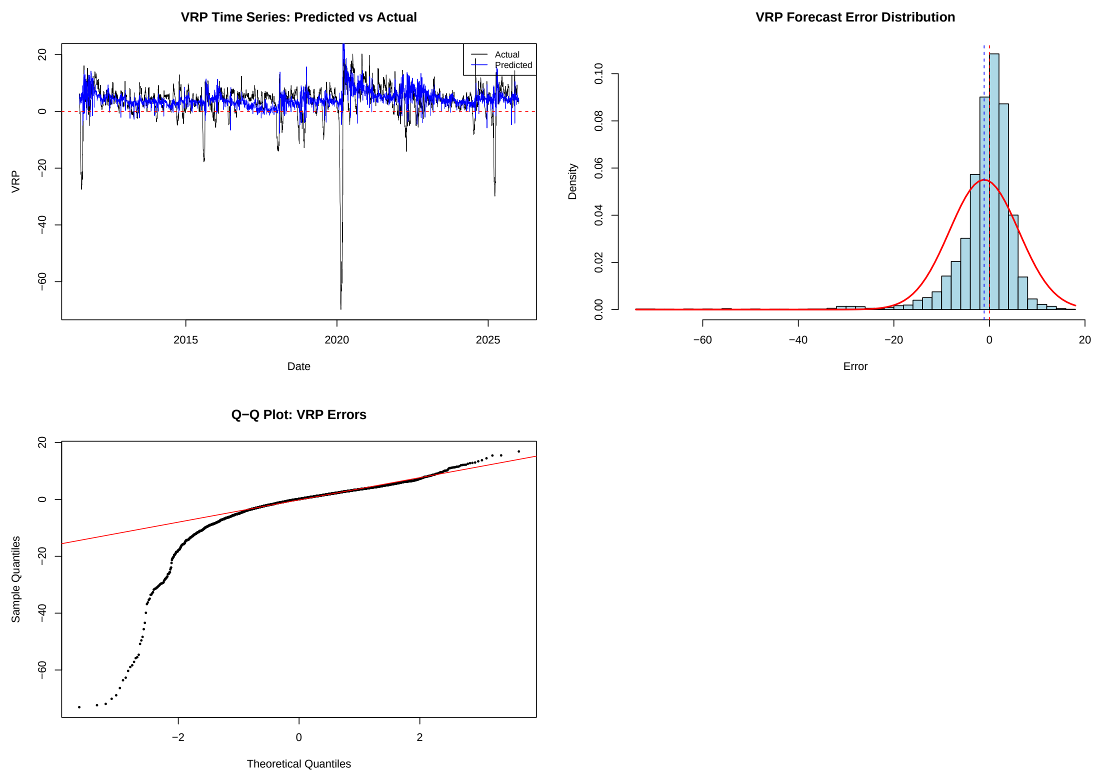

Figure 5.16: Actual versus predicted variance risk premium over the test period ($N = 3{,}024$).

Table 5.10 presents the descriptive statistics of the VRP forecast components and errors.

Table 5.10: Descriptive statistics of VRP forecast components ($N = 3{,}024$).

Variable	Mean	Std. Dev.	Skewness	Kurtosis	Min	Max
VRP Predicted	4.64	3.22	1.78	14.56	−10.27	37.01
VRP Actual	3.36	7.33	−4.51	37.13	−69.82	20.27
VRP Forecast Error	−1.27	7.36	−4.17	32.26	−73.38	17.27
VIX Predicted	18.08	6.77	2.72	16.89	9.56	79.15
VIX Actual	17.98	6.96	2.70	16.83	9.14	82.69
RV Predicted	13.44	4.75	2.96	15.14	8.81	43.18
RV Actual	14.61	9.63	4.03	28.40	3.72	93.55

Table 5.11 presents the overall VRP forecast metrics and compares the hierarchical model against simpler benchmarks.

Table 5.11: VRP forecast accuracy: hierarchical model versus benchmarks.

Model	**RMSE**
AR(1) VRP	2.099
Random Walk (VRP_{t-1})	2.121
XGBoost (VIX + RV)	7.463
Historical Mean	7.325
Zero VRP	8.060

5.7.1 Error Decomposition

Table 5.12 decomposes the VRP forecast error. The RV component dominates (108.1% of variance; RMSE = 7.65 vs. 2.46 for VIX), confirming that the 22-day-ahead RV prediction is the binding constraint. The negative covariance (-19.3%) provides a partial natural hedge. The RV contribution exceeds 100% because the negative covariance between VIX and RV forecast errors (19.3%) acts as a partial offset. This means the individual component variances sum to more than the total VRP error variance.

Table 5.12: VRP forecast error variance decomposition.

Source	**Variance**	**% of VRP Var.**	**RMSE**
VIX Forecast Error	6.03	11.1%	2.455
RV Forecast Error	58.50	108.1%	7.648
Covariance Term	−10.42	−19.3%	—

5.7.2 Mincer–Zarnowitz Efficiency

Table 5.13 presents MZ results across the hierarchy. The VIX forecast is near-efficient ($\hat{\beta} = 0.963$, $R^2 = 0.877$). The RV component is over-responsive ($\hat{\beta} = 1.256$, $R^2 = 0.385$). The VRP forecast is severely attenuated ($\hat{\beta} = 0.479$, $R^2 = 0.044$; HAC $t = -4.79$, $p < 0.001$), failing to capture the full amplitude of VRP swings. This calibration failure despite good component forecasts arises from the horizon mismatch and nonlinear error interactions.

Table 5.13: Mincer–Zarnowitz regression results for VRP and component forecasts.

Forecast	$\hat{\alpha}$	$\hat{\beta}$	R^2	**Interpretation**
VRP	1.142	0.479	0.044	Severely attenuated
VIX	0.564	0.963	0.877	Near-efficient
RV	−2.269	1.256	0.385	Over-responsive

5.7.3 Trading Signal Architecture

Given the poor VRP point accuracy (MZ $R^2 = 0.044$) but strong sign accuracy (81.5%), a multi-layered signal extracts directional and regime information through an AND-gate combining two components.

Signal 1: VRP Z-Score Regime Classification. The VRP forecast is normalised via a 252-day rolling z-score $z_t^{\mathrm{VRP}} = (\widehat{\mathrm{VRP}}_t - \bar{\mu}_{t,252})/\bar{\sigma}_{t,252}$ and mapped to five regimes:

Table 5.14: VRP z-score regime classification and position sizing.

Regime	Z-Score Range	Scalar	Interpretation
Extremely Rich	$z > +1.5$	0.00	Tail risk imminent; no trade
Rich	$+0.5 < z \leq +1.5$	0.33	Premium rich for a reason; minimal size
Normal	$0 < z \leq +0.5$	1.00	Premium near average; full size
Lean	$-0.5 < z \leq 0$	0.33	Premium below average; minimal size
Inverted	$z \leq -0.5$	0.00	Premium compressed; no trade

Signal 2: Forecast Uncertainty Regime Filter. The 95% VIX prediction interval width, normalised via a 252-day rolling z-score, proxies forecast uncertainty:

Table 5.15: Forecast uncertainty regime classification.

Regime	Z-Score Range	Multiplier	Interpretation
Stable	$z < +0.5$	1.00	Normal uncertainty; no adjustment
Elevated	$+0.5 \leq z < +1.0$	0.75	Elevated uncertainty; reduce exposure
Stressed	$+1.0 \leq z < +1.5$	0.50	High uncertainty; halve exposure
Crisis	$z \geq +1.5$	0.00	Extreme uncertainty; no trade

When prediction intervals are abnormally wide, the VRP forecast is unreliable regardless of its point estimate.

Composite AND-Gate. A trade requires: (i) $\widehat{\mathrm{VRP}}_t > 0$, (ii) VRP scalar $s_1 > 0$ (not Extremely Rich or Inverted), and (iii) regime multiplier $s_2 > 0$ (not Crisis). The composite scalar $s_{\mathrm{composite}} = s_1 \cdot s_2$ produces an inverted-U position sizing profile. Both extremes of the VRP distribution are excluded. Extremely rich VRP is not treated as a stronger buy signal but as a warning, since historically elevated premia have preceded tail events such as March 2020 and Volmageddon, where the VRP was rich precisely because the market was pricing genuine crash risk. Supplementary adjustments reduce position size by 25% when the VIX classification model predicts $P(\Delta\mathrm{VIX} > 0) > 0.6$.

5.7.4 Bull Put Spread Implementation

All strategies considered had to be buy and hold, and monthly trades due to data constraints. Based on this it was decided that the signal would be implemented via delta-targeted bull put spreads on the S&P 500, providing defined-risk VRP exposure. The short put targets $\delta = -0.16$ ($\approx 1\sigma$ OTM), the long put targets $\delta = -0.05$, with maximum loss bounded by:

$$\text{Max Loss} = (K_1 - K_2) - \text{Net Premium} \tag{5.1}$$

Position sizing uses fractional Kelly with the composite scalar modulating the risk budget: $\text{Risk Budget}_t = \text{NAV}_t \times r_{\text{max}} \times s_{\text{composite},t}$. Execution assumes mid-price fills with 50% half-spread slippage; positions are held to expiry. Due to low risk budget in proportion to total NAV, margin requirements from brokers would be met.

5.7.5 Backtest Results

The system is backtested from June 2011 to February 2026 with $1,000,000 initial capital (Table 5.16).

Table 5.16: Backtest performance summary (June 2011 – February 2026).

Metric	**Value**
Initial Capital	$1,000,000
Final NAV	$1,530,436
Total Return	53.0%
Total Trades	68
Winning Trades	64
Losing Trades	4
Hit Rate	94.1%
Average Win	$10,441
Average Loss	−$34,444
Largest Win	$22,200
Largest Loss	−$77,655
Sharpe	1.24

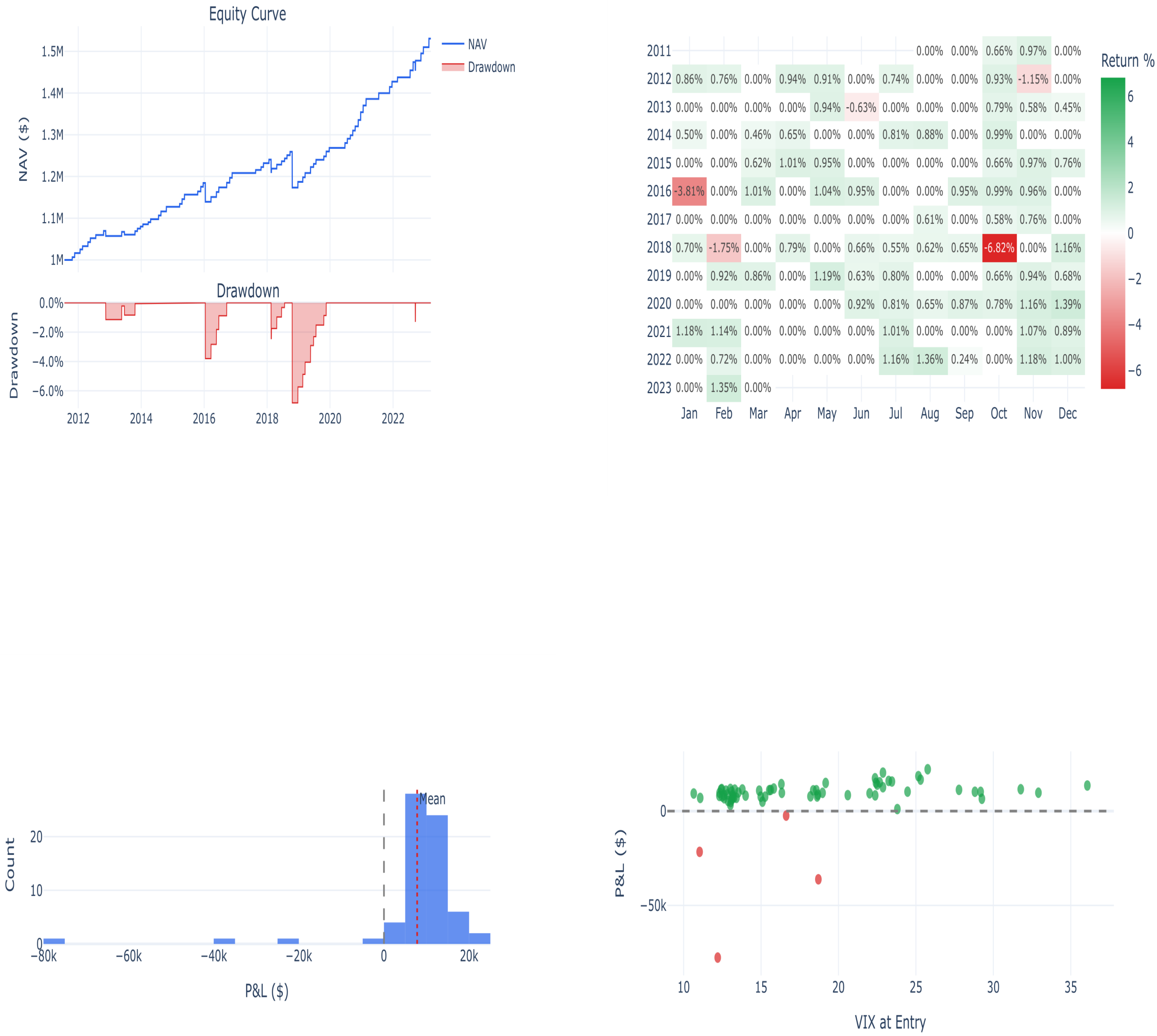

Figure 5.17: Performance summary of the VRP signal-gated SPX put spread strategy.

The strategy achieves a 53.0% total return with a 94.1% hit rate across 68 trades. The average loss ($34,444) is 3.3× the average win ($10,441), reflecting the negative skewness of short volatility returns, but the high hit rate produces positive expected value. The 94.1% hit rate exceeds the 81.5% VRP sign accuracy because the AND-gate filters high-uncertainty periods, the regime classification avoids VRP compression, and delta-targeted construction provides an OTM buffer. Table 5.17 presents annual performance.

Table 5.17: Annual P&L breakdown.

Year	Trades	P&L ($)	Cumulative ($)
2014	8	81,300	81,300
2015	8	35,320	116,620
2016	8	67,100	183,720
2017	9	90,440	274,160
2018	5	−44,904	229,256
2019	7	83,300	312,556
2020	3	41,100	353,656
2021	6	56,880	410,536
2022	4	22,500	433,036
2023	5	50,200	483,236
2024	4	33,200	516,436
2025	1	14,000	530,436

The strategy is profitable in every year except 2018 (Volmageddon), where the $44,904 loss (4.5% of capital) was contained by the defined-risk spread structure. In 2020, the regime filter correctly identified March as CRISIS, suspending trading; the three trades executed during the calmer recovery captured the elevated post-crash VRP.

The VRP z-score is the primary filter (38 rejections), followed by build failures (21), regime crisis (15), and negative VRP (12). The AND-gate is critical: a VRP-only signal would have entered several high-uncertainty episodes that produced losses. The four losing trades occur when S&P 500 declines breach the short strike, with the largest loss ($77,655, Volmageddon) representing 7.0% of NAV — contained by the long put leg.

Chapter 6

Conclusion

6.1 Summary of Findings

This dissertation developed a hierarchical machine learning framework for VRP estimation, decomposing the problem into VIX prediction (AR(1)+XGBoost), realised volatility prediction (XGBoost), and hierarchical VRP construction.

For VIX forecasting (RQ1), the AR(1)+XGBoost model achieves RMSE = 1.886 (Theil $U = 0.995$), comparable to GARCH(1,1) (RMSE = 1.884) and substantially outperforming HAR (RMSE = 2.397). Directional accuracy reaches 57.8% (AUC-ROC = 0.600). For RV forecasting (RQ2), XGBoost achieves RMSE = 7.49, outperforming the VIX benchmark (RMSE = 8.34), with MZ $R^2 = 0.385$. For VRP estimation (RQ3), the hierarchical forecast achieves 81.5% sign accuracy; the backtest produces a 94.1% hit rate, Sharpe ratio of 1.24, and 53.0% total return across 68 trades. SHAP analysis (RQ4) reveals regime-dependent feature importance, with cross-asset and volatility clustering features gaining prominence in high-volatility environments.

6.2 Contributions

This dissertation makes five main contributions. First, it proposes and evaluates the AR(1)-residual XGBoost framing, which isolates the nonlinear signal from the dominant autoregressive component and focuses the machine learning model on the portion of VIX dynamics that traditional models fail to capture. Second, it develops a hierarchical VRP prediction architecture with formal error decomposition, enabling transparent attribution of VRP forecast errors to their component sources and identifying RV forecasting as the binding constraint. Third, it designs and backtests a multi-layered trading signal architecture, combining VRP z-score regime classification with forecast uncertainty filtering through an AND-gate. Within the monthly buy-and-hold implementation constraints imposed by the data, the strategy achieved a 94.1% hit rate and a Sharpe ratio of 1.24 across 68 trades. Fourth, it provides comprehensive regime-conditional evaluation, demonstrating that XGBoost's advantage over benchmarks varies systematically with the volatility environment. Fifth, it applies SHAP-based interpretability analysis across regimes, providing granular insight into the feature-level drivers of volatility forecast accuracy in different market conditions.

6.2.1 The Architecture as Contribution

Independent of the specific forecasting results, the hierarchical decomposition represents a methodological contribution that persists regardless of the machine learning method employed. Most existing studies forecast VRP as a monolithic quantity, training a single model on the VRP time series. The hierarchical approach offers several structural advantages:

1. **Transparency.** The error decomposition (Table 5.12) immediately identifies the binding constraint (RV forecasting), directing future modelling effort toward the component with the greatest marginal return. A monolithic VRP model would not provide this diagnostic.

2. **Modularity.** Each component can be upgraded independently. The VIX model could be replaced with a neural network, the RV model with a HAR-LSTM hybrid, or the variance estimator with a high-frequency realised measure, without affecting the other components.

3. **Interpretability.** The intermediate forecasts (VIX and RV separately) have independent economic value for risk management and portfolio construction, beyond their role in VRP estimation.

4. **Horizon alignment.** The framework makes the horizon mismatch between VIX (one-step-ahead) and RV (22-step-ahead) forecasts explicit, rather than burying it within a single model. This transparency motivates research into multi-horizon forecasting architectures.

6.3 Future Work

Several directions for future research emerge from this work.

First, the incorporation of high-frequency intraday data would enable the construction of substantially more precise realised volatility estimates, likely improving the RV forecasting component and consequently the VRP prediction accuracy.

Second, incorporating additional data sources beyond historical VIX and SPX may lead to increased model performance. Other similar studies use treasury data, commodity prices, FX rates and Natural Language Processing (NLP) for constructing additional features.

Third, extending the framework to a cross-sectional dimension, such as estimating VRP for individual equities, sector ETFs, or commodity indices.

Fourth, deep learning extensions. Long Short-Term Memory (LSTM) networks can capture long-range temporal dependencies more explicitly than XGBoost, while the Temporal Fusion Transformer architecture combines attention mechanisms with multi-horizon forecasting. These models could replace or augment the XGBoost component within the hierarchical framework.

Fifth, ensemble methods such as stacking XGBoost with GARCH conditional variance as an additional feature, or combining multiple ML models via weighted averaging could further improve forecast accuracy by exploiting the complementary strengths of different modelling approaches.

Finally, the computational aspects of the framework provide a natural bridge to high-performance computing research. GPU-accelerated XGBoost training, parallel Bayesian optimisation across cores, and rolling-window model re-estimation over long sample periods are all concrete applications of HPC techniques to financial machine learning that could be explored in subsequent work.

6.4 Limitations

Several limitations should be acknowledged. The VRP forecast is dominated by the RV component (108.1% of error variance), meaning improvements to RV forecasting, via high-frequency data or deep learning, represent the most impactful avenue for future work. Transaction costs are approximated rather than calibrated to live execution data. Due to computational constraints the XGBoost models are trained on a fixed training window (1990–2011) and applied without retraining over a 15-year out-of-sample period. Walk-forward retraining with an expanding or rolling window might better account for structural shifts in VIX dynamics.

Bibliography

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96(453):42–55, 2001.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003.

Ole E. Barndorff-Nielsen and Neil Shephard. Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2):253–280, 2002.

Geert Bekaert and Marie Hoerova. The VIX, the variance premium and stock market volatility. *Journal of Econometrics*, 183(2):181–192, 2014.

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

Tim Bollerslev and Viktor Todorov. Tails, fears, and risk premia. *Journal of Finance*, 66(6):2165–2211, 2011.

Tim Bollerslev, George Tauchen, and Hao Zhou. Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11):4463–4492, 2009.

Tim Bollerslev, Andrew J. Patton, and Rogier Quaedvlieg. Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192 (1):1–18, 2016.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Wadsworth, 1984.

Mark Britten-Jones and Anthony Neuberger. Option prices, implied price processes, and stochastic volatility. *Journal of Finance*, 55(2):839–866, 2000.

Andrea Bucci. Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531, 2020.

John Y. Campbell and John H. Cochrane. By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107 (2):205–251, 1999.

Peter Carr and Liuren Wu. Variance risk premiums. *Review of Financial Studies*, 22 (3):1311–1341, 2009.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. pages 785–794, 2016.

Bent J. Christensen and Nagpurnanand R. Prabhala. The relation between implied and realized volatility. *Journal of Financial Economics*, 50(2):125–150, 1998.

Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.

Kresimir Demeterfi, Emanuel Derman, Michael Kamal, and Joseph Zou. More than you ever wanted to know about volatility swaps. *Goldman Sachs Quantitative Strategies Research Notes*, 1999.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982.

Marcelo Fernandes, Marcelo C. Medeiros, and Marcel Scharth. Modeling and predicting the CBOE market volatility index. *Journal of Banking & Finance*, 40:1–10, 2014.

Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

Jerome H. Friedman. Stochastic gradient boosting. *Computational Statistics & Data Analysis*, 38(4):367–378, 2002.

Mark B. Garman and Michael J. Klass. On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1):67–78, 1980.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Peter Reinhard Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, 2nd edition, 2009.

Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343, 1993.

Fabian Hollstein and Chardin Wese Simen. Variance risk: A bird's eye view. *Journal of Econometrics*, 215(2):518–535, 2020.

Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An Introduction to Statistical Learning*. Springer, 2013.

Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13(4): 455–492, 1998.

Marcos López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. 30:4765–4774, 2017.

Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.

Jacob A. Mincer and Victor Zarnowitz. The evaluation of economic forecasts. In *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, pages 3–46. NBER, 1969.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55 (3):703–708, 1987.

Michael Parkinson. The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1):61–65, 1980.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2024.

Carl Edward Rasmussen and Christopher K.I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

L. Chris G. Rogers and Stephen E. Satchell. Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4):504–512, 1991.

Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. volume 104, pages 148–175. 2016.

Lloyd S. Shapley. A value for *n*-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games*, volume 2, pages 307–317. Princeton University Press, 1953.

Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. Practical Bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems*, volume 25, pages 2951–2959, 2012.

Robert E. Whaley. Derivatives on market volatility: Hedging tools long overdue. *Journal of Derivatives*, 1(1):71–84, 1993.

Dennis Yang and Qiang Zhang. Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3):477–492, 2000.

Hao Zhou. Variance risk premia, asset predictability puzzles, and macroeconomic uncertainty. *Annual Review of Financial Economics*, 10:481–497, 2018.

Appendix A

Hyperparameter Search Spaces and Optimal Values

This appendix reports the full hyperparameter search spaces and optimal values selected by Bayesian optimisation for each of the four XGBoost models: VIX regression, VIX classification, RV regression, and RV classification.

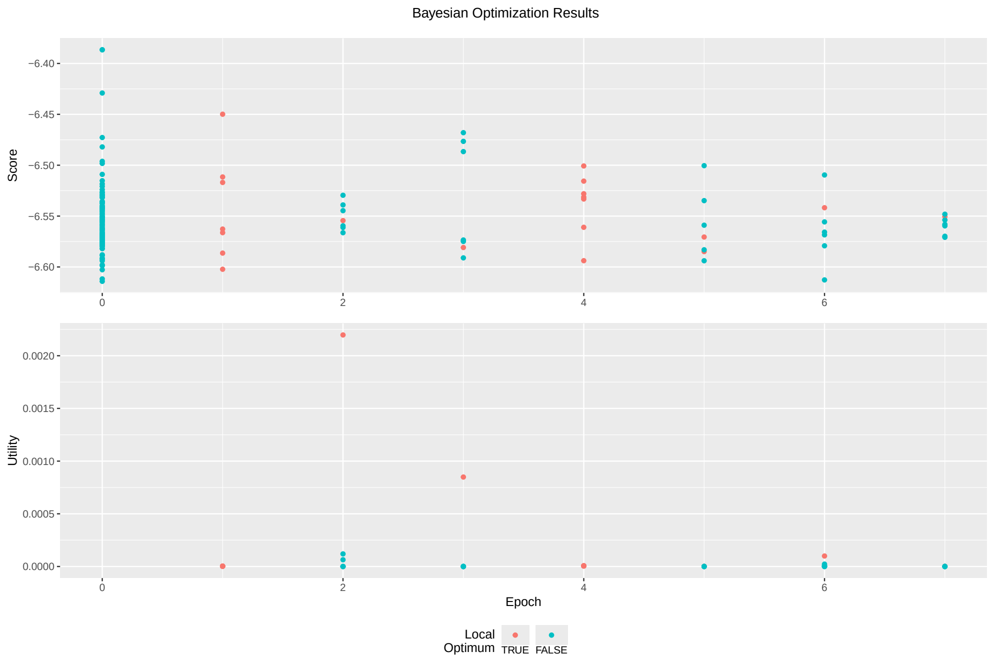

Figure A.1: Bayesian optimisation convergence for the VIX regression (AR(1) residual) model.

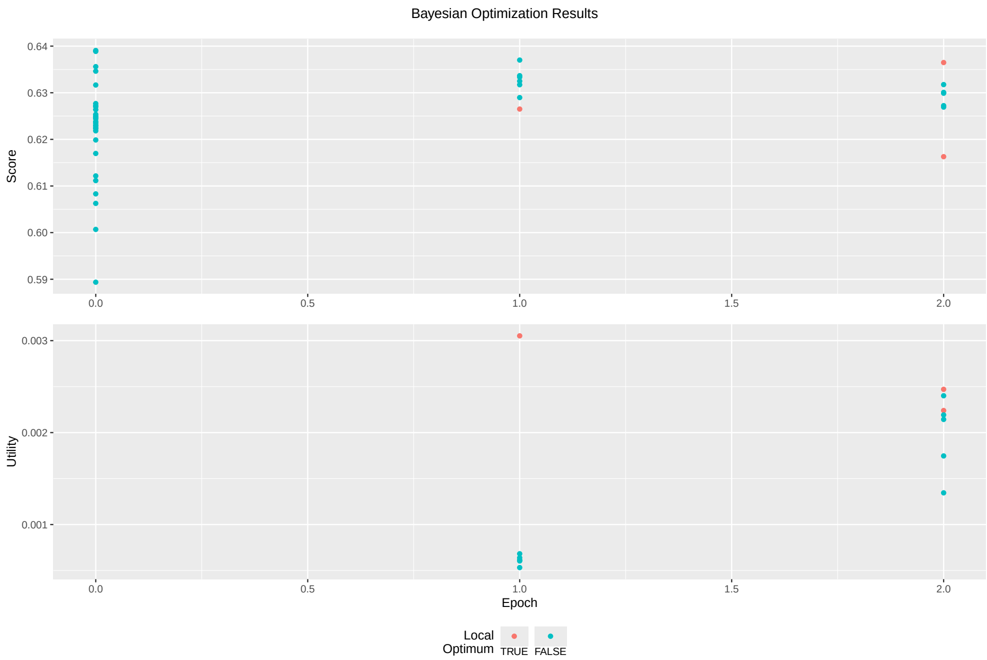

Figure A.2: Bayesian optimisation convergence for the VIX directional classification model.

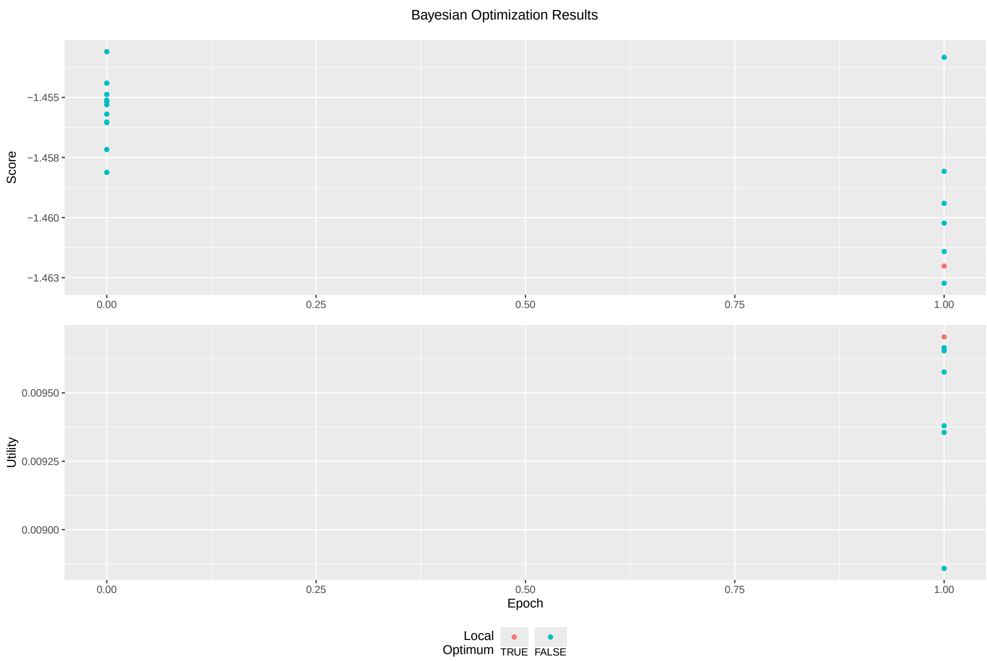

Figure A.3: Bayesian optimisation convergence for the RV regression model (v2).

Appendix B

Reproducibility

Full reproducibility information is provided including: R package versions, session information, random seeds used for all stochastic processes (Bayesian optimisation, subsampling), and the complete script execution order.

Following key packages and their roles:

Table B.1: Key R packages used in the analysis.

Package	**Purpose**
`xgboost`	Gradient boosting model training and prediction
`ParBayesianOptimization`	Bayesian hyperparameter optimisation
`SHAPforxgboost`	SHAP value computation and visualisation
`rugarch`	GARCH model estimation (GARCH, EGARCH, GJR, TGARCH)
`quantmod`	Financial data download and OHLC manipulation
`forecast`	HAR model estimation and time series utilities
`doParallel`	Parallel computation backend
`data.table`	High-performance data manipulation
`ggplot2`	Statistical graphics
`fracdiff`	Fractional differencing

Script execution order:

1. `Setup.R` — Package loading and environment configuration
2. `Data Download.R` — VIX and SPX data retrieval from Yahoo Finance
3. `Data Prep and Exploration.R` — Data cleaning and initial EDA
4. `CBOE VIX Construction.R` — VIX replication and validation
5. `HAR GARCH Models.R` — Benchmark econometric model estimation
6. `VIX Feature Engineering.R` — Feature engineering for VIX forecasting
7. `XGBOOST Classification.R` — XGBoost VIX classification model
8. `XGBOOST Regression.R` — XGBoost VIX regression model
9. `Total Model Assessment.R` — Comprehensive model comparison
10. `Realised Volatility Construction.R` — OHLC-based RV estimator construction
11. `Realised Volatility Exploratory Data Analysis.R` — RV stylised facts

12. `RV Feature Engineering.R` — Feature engineering for realised volatility forecasting
13. `RV XGBOOST Regression.R` — XGBoost RV regression model
14. `RV XGBOOST Classification.R` — XGBoost RV classification model
15. `Volatility Risk Premium Analysis.R` — Volatility risk premium estimation and analysis

Appendix C

Table of Notation

Symbol	Description
S_t	Asset price at time t
σ_t	Instantaneous volatility at time t
σ_t^2	Instantaneous variance at time t
VIX_t	CBOE Volatility Index at time t
RV_t	Realised volatility (22-day rolling, annualised)
VRP_t	Variance risk premium: $\text{VIX}_t - \text{RV}_{t,t+22}$
$\mathbb{P}$	Physical (statistical) probability measure
$\mathbb{Q}$	Risk-neutral probability measure
$\mathbb{E}^{\mathbb{P}}[\cdot]$, $\mathbb{E}^{\mathbb{Q}}[\cdot]$	Expectations under physical and risk-neutral measures
M_t	Pricing kernel (stochastic discount factor)
W_t	Standard Brownian motion
r_t	Log return: $\log(C_t/C_{t-1})$
O_t, H_t, L_t, C_t	Open, High, Low, Close prices on day t
$\hat{\sigma}^2_{\text{CC},t}$	Close-to-close daily variance estimate
$\hat{\sigma}^2_{\text{P},t}$	Parkinson daily variance estimate
$\hat{\sigma}^2_{\text{GK},t}$	Garman–Klass daily variance estimate
$\hat{\sigma}^2_{\text{RS},t}$	Rogers–Satchell daily variance estimate
$\hat{\sigma}^2_{\text{YZ},t}$	Yang–Zhang daily variance estimate
d	Fractional integration (long-memory) parameter
ϕ	AR(1) coefficient
$f_k(\mathbf{x})$	k-th tree in the XGBoost ensemble
η	Learning rate (shrinkage parameter)
K	Number of boosting iterations
T_k	Number of leaves in tree k
$\mathbf{w}_k$	Leaf weight vector for tree k
g_i, h_i	First and second derivatives of loss function
γ	Minimum split gain (XGBoost regularisation)
λ	L_2 regularisation parameter

Symbol	**Description**
α	L_1 regularisation parameter
$\phi_j(i)$	SHAP value of feature j for instance i
ϕ_0	SHAP base value (mean prediction)
$\mathbb{1}\{\cdot\}$	Indicator function
RMSE	Root mean squared error
MAE	Mean absolute error
MAPE	Mean absolute percentage error
QLIKE	Quasi-likelihood loss function
DM	Diebold–Mariano test statistic
CW	Clark–West adjusted test statistic
PT	Pesaran–Timmermann test statistic
MZ	Mincer–Zarnowitz regression